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Infectious Diseases and the Kidney

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CHAPTER OUTLINE

KEY POINTS

- **Infections can lead to the development and worsening of a range of kidney diseases including acute kidney injury (AKI), tubular disorders, chronic kidney disease, and urinary tract abnormalities.**
- **Mechanisms of kidney injury due to infections include direct invasion, tissue inflammation, toxin release, vascular supply interference, and urinary tract obstruction.**
- **Tropical areas, characterized by their warm and humid climate, provide a conducive environment for the growth of pathogenic microorganisms and vectors, many of which involve the kidneys.**
- **Factors such as overcrowded living conditions, occupational risks, poor sanitation, restricted clean water access, malnutrition, and limited health care availability render economically disadvantaged communities more susceptible to infection-associated kidney disease.**
- **The COVID-19 pandemic has disproportionately impacted individuals with kidney disease, resulting in higher mortality rates, AKI, and the emergence of new-onset glomerular diseases.**
- **The spectrum of malaria-associated kidney injury has changed over time, with *Plasmodium vivax* and *Plasmodium knowlesi* emerging as important causes of AKI. In contrast, quartan malarial nephropathy has disappeared from Africa.**
- **Chyluria due to filarial involvement of the kidneys and urinary tract can be mistaken for nephrotic syndrome.**
- **Fungal infections of the kidney, like aspergillosis and mucormycosis, can develop in immunocompetent individuals.**

Kidney diseases are typically placed in the general category of noncommunicable diseases.[1] Infections, however, can result in the development of a diverse array of kidney diseases, encompassing acute kidney injury (AKI), tubular disorders, acute and chronic interstitial nephritis, nephrotic and nephritic syndromes, chronic kidney disease (CKD), and urinary tract abnormalities.[2]

A comprehensive understanding of infections that give rise to kidney diseases is important for prompt diagnosis and timely commencement of interventions, thereby preventing or mitigating the damage caused by infections. A link between infections and kidney disease can be suspected quickly when the presentation is acute and the exposure is temporally proximate. Sometimes, the presentation may be related to an obscure or remote (temporally or geographically) infection. Hence a thorough knowledge of the local epidemiology of infections, detailed history taking including travel history and potential exposures, investigation of current and past medical records, a complete physical examination, and judicious utilization of microbiologic studies are imperative for an accurate diagnosis.

A wide range of microbial species including viruses, bacteria, mycobacteria, fungi, and protozoa can cause kidney involvement.[2] The prevalence and distribution of infectious diseases vary globally with geography and populations and change dynamically over time. The epidemiology of infection-related kidney disease mirrors the relative prevalence of infections in different parts of the world. The unique climactic conditions (warmth and humidity), abundance of water, limited seasonal variation, and rich biodiversity in the tropical areas make them particularly conducive to the growth of disease-causing microorganisms and vectors, giving rise to infections that involve the kidneys. Poor people living in these regions experience additional risk factors (overcrowded housing, occupational hazards, inadequate sanitation, limited access to clean water, malnutrition, and limited health care access) and are therefore particularly vulnerable. [3] It is crucial to recognize the impact of infections on the development and/or progression of kidney diseases in these regions. This impact is significant in terms of both direct effects and indirect consequences, leading to considerable morbidity and mortality. The ability to promptly identify an infection depends on the level of clinical suspicion and access to appropriate diagnostic tests. The former is complicated by a considerable overlap in the clinical signs and symptoms[4] (Table 59.1), whereas the latter is influenced by system-level factors, especially in low-resource countries and among disadvantaged populations in high-income countries.[5]

Table 59.1 Common Differential Diagnosis of Tropical Acute Febrile Illness Associated with Acute Kidney Injury

Clinical Picture	Differential Diagnosis
Fever + jaundice	Leptospirosis, malaria, dengue, hantavirus, rickettsiosis, acute hepatitis
Biphasic fever + conjunctival suffusion + thrombocytopenia + transaminitis	Leptospirosis
Continuous fever + severe respiratory symptoms leading to ARDS	Hantavirus
Fever + severe myalgia + thrombocytopenia + acalculous cholecystitis	Dengue
Fever + maculopapular rash + eschar	Scrub typhus
Fever + splenomegaly + thrombocytopenia	Malaria
Fever + exposure to unpasteurized milk products	Brucellosis
Fever + diarrhea	Bacterial or viral gastroenteritis

AKI, acute kidney injury; *ARDS,* acute respiratory distress syndrome; *CA-AKI,* community-acquired acute kidney injury.
From Burdmann EA, Jha V. Acute kidney injury due to tropical infectious diseases and animal venoms: a tale of 2 continents. *Kidney Int*. 2017;91(5): 1033–1046.

Many countries have made large gains in infection control and achieved substantial reductions in infection-related morbidity and mortality through public health measures like mass vaccination, improved sanitation, and the availability of better diagnostic techniques and therapeutic agents. [6] However, these gains are not distributed equitably, with the socioeconomically disadvantaged populations shouldering the largest residual burden.[7] Further, emerging challenges such as global environmental change and the degradation of biodiversity pose the risk of undoing those gains and even expanding the geography vulnerable to infections and associated kidney diseases.[8] Finally, the indiscriminate utilization of antimicrobials has presented the predicament of the emergence of resistant organisms that effectively limit our ability to manage infections, such as urinary tract infections. [9]

GENERAL PRINCIPLES OF KIDNEY INVOLVEMENT IN INFECTIONS

Infections can induce kidney injury through several mechanisms, which can be broadly categorized into *organism-related,* such as direct invasion, tissue inflammation, elaboration of toxins, interference with vascular supply, and urinary tract obstruction; *host-related,* such as systemic inflammation response syndrome leading to hemodynamic alterations, immune-mediated injury, activation of coagulation or complement pathways; or *tissue inflammation.*[2,10–15] They are often compounded by other factors, such as the use of nephrotoxic drugs, volume (including blood) loss, rhabdomyolysis, and hemolysis[16,17] (Fig. 59.1).

For example, bacteria can directly invade the renal parenchyma, cause an immune response, damage the capillary endothelium, cause a humoral reaction, elicit granuloma formation [like in leprosy or tuberculosis (TB)], or activate the complement pathway.[11,13–20] Bacterial pathogens can initiate activation of innate immunity or macrophages, which in turn stimulate T cell and B cell responses.[21] This can result in immune complex–mediated glomerular injury, as seen in infectious-related glomerulonephritis (GN) or interstitial nephritis.[22] The development of an inflammatory response, such as during leptospirosis, can also lead to kidney injury. Viral infections primarily impact the kidneys through the activation of the immune system, as in the case of dengue infection.[23] Immune complex formation can result in glomerular disease, as observed in hepatitis C and hepatitis B virus infections.[11,22] Urinary tract obstruction can develop following infections due to certain fungi or a nematode like *Schistosoma hematobium.*[10] Endothelial activation is one of the key mechanisms of organ and tissue injury in severe malaria.[24] Scrub typhus can result in poor renal perfusion, causing systemic vasculitis and increased vascular permeability. Angioinvasive fungi such as mucor can lead to thrombosis of large vessels and cause renal infarction. [25]

This chapter discusses kidney involvement in various infections (Table 59.2) including malaria, leptospirosis, dengue fever, scrub typhus, mycobacterial diseases, neglected tropical infections, diarrheal diseases, and coronavirus disease 2019 (COVID-19). Infections that are especially prevalent in one particular region of the world are discussed in regional chapters elsewhere in this book.

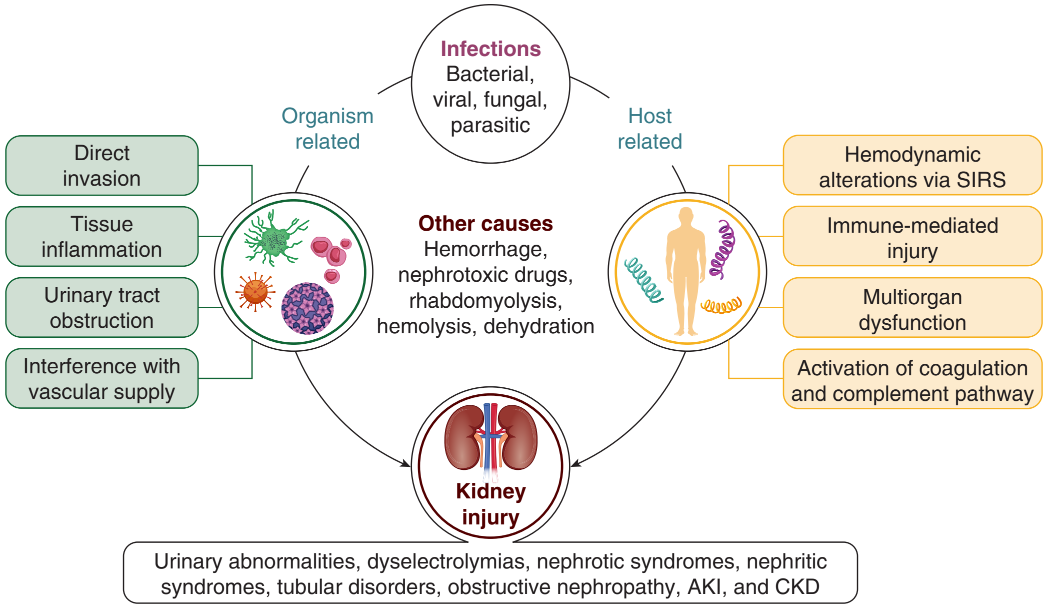

Fig. 59.1 Mechanism of kidney involvement in infections. *AKI,* Acute kidney injury; *CKD,* chronic kidney disease; *SIRS,* systemic inflammatory response syndrome.

This chapter does not cover infection-associated GN or infections that primarily affect immunosuppressed kidney transplant recipients (KTRs) which are discussed in more detail respectively in Chapters 34 and 69. Urinary tract infections are discussed in detail in Chapter 38. The primary emphasis is on examining the underlying mechanisms of pathogenesis, identifying characteristic features, and discussing preventive measures and management strategies.

VIRAL INFECTIONS

COVID-19

Coronavirus disease 2019 (COVID-19) is caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), an enveloped positive-stranded ribonucleic acid (RNA) virus belonging to the family Coronaviridae, the order Nidovirales, and the genus *Coronavirus*. Phylogenetic analysis shows it to be closely related (96.3%) to the bat coronavirus RaTG13.[26] The high level of contagiousness led to its rapid global spread since the identification of the initial outbreak in Wuhan, China, in December 2019, reaching the status of a pandemic in just 2 months and causing close to 7 million deaths over the next couple of years.[27] Since the initial identification and sequencing, SARS-CoV-2 has continued to evolve, with various strains of different transmissibility, virulence, and potential for immune escape. The transmission of COVID-19 primarily occurs through respiratory droplets.

CLINICAL MANIFESTATIONS

Disease manifestations are determined by both virus and host factors. The disease remains asymptomatic in 10% to 60% of cases, while others develop a combination of sore throat, cough, myalgia, fever, dyspnea, fatigue, headache, anosmia, and ageusia. Although COVID-19 primarily affects the respiratory system, the virus can involve multiple organs including kidneys.

Kidney involvement in patients with COVID-19 may manifest as isolated urinary abnormalities, such as hematuria and/or proteinuria, AKI, or GN. In initial studies, the incidence of AKI ranged from 32% to 46%, and it has varied in different studies depending on the severity of the disease and the need for hospitalization and/or intensive care.[28–30]

In a retrospective, observational study comprising a cohort of 3993 hospitalized patients diagnosed with COVID-19, AKI occurred in 1835 (46%) patients and 19% of those with AKI needed dialysis. Stage 3 AKI was present in 42% of cases. Of those hospitalized in the intensive care unit (ICU), 76% had AKI. Among the cohort of patients with AKI, proteinuria, hematuria, and leukocyturia were found in 84%, 81%, and 60%, respectively.[28]

Patients with CKD, especially those on dialysis, those receiving immunosuppressive therapy, and KTRs, are at increased risk of getting infected with COVID-19, developing more severe disease, requiring hospitalization, and being at a greater risk of adverse outcomes including mortality.[31–34] Analysis of the NHS Digital Trusted Research Environment for England (NHSD TRE) database from March 2020 to March 2021 showed that after adjusting for age, gender, COVID-19 vaccinations, and positive COVID-19 test results with adequate matching, patients on dialysis or transplantation had the highest relative risk of 1-year all-cause death associated with SARS-CoV-2 infection (prevalent CKD: relative risk, 1.70; 95% confidence interval [CI], 1.67–1.73). This vulnerability is attributed to the attenuated activation of innate and adaptive immune systems in patients with CKD.[35] Patients with CKD, especially those on dialysis, also show delayed viral clearance.[36,37]

The COVID-19 pandemic has had many repercussions on the delivery of care to patients with kidney disease. These effects included interruptions in dialysis delivery, delayed kidney biopsies, prolonged waiting times for arteriovenous fistula surgery and salvage procedures, disruption of kidney transplant services, interruptions in the supply chain of lifesaving drugs and disposables, and psychological distress to patients, family members, and caregivers including health care professionals.[38–42] Studies have also suggested a significant association between the COVID-19 epidemic and high perceived stress among KTRs.[43]

Nephrotic or nephrotic-nephritic presentations due to a range of glomerular diseases (see later) have been reported in COVID-19. Further, relapses of nephrotic syndrome in children and recurrences of antiglomerular basement disease have also been documented.[44,45] In addition, a second or third dose of the COVID-19 vaccine was associated with a higher relative but a low absolute increased risk of relapse in patients with glomerular disease.[46]

PATHOGENESIS

The development of kidney injury in COVID-19 involves multiple mechanisms including direct cytotoxicity, hypercoagulation, virus-induced cytokine storms, activation of the angiotensin II pathway, complement dysregulation, and microangiopathy. [47]

The virus can enter kidney cells through the interaction between the receptor-binding domain (RBD) of the spike protein of the virus and the angiotensin-converting enzyme II (ACE2) present on the surface of proximal tubular cells and podocytes. Subsequently, the spike protein undergoes proteolytic cleavage mediated by the transmembrane protease serine 2 (TMPRSS2).[48] Podocytes and proximal tubular cells expressing the *ACE2* gene have been identified as key host cells for SARS-CoV-2. SARS-CoV-2 can also infiltrate host cells via CD147, a transmembrane glycoprotein widely expressed on proximal tubular cells.[49,50] After entering cells, SARS-CoV-2 releases its constituents, replicates, and infects other cells.[51] Kidney macrophages interact with viruses and stimulate phagocyte and chemokine signaling.[52] The cytopathic impact of the SARS-CoV-2 virus can directly harm renal tubular cells during infection and replication, triggering an intricate immunologic response.[52] Activated lymphocytes in the renal interstitium initiate renal cell destruction, leading to the release of proinflammatory cytokines such as perforin and granulysin.[47] The resulting cytokine storm may contribute to AKI. Interleukin-6, a key cytokine, induces the secretion of other proinflammatory chemokines and cytokines and increases kidney endothelial vascular permeability, contributing to microcirculatory dysfunction.[53] Natural killer cells and $CD4^+$ T cells infiltrate the tubular interstitium and secrete proinflammatory cytokines, causing tubular damage, fibrosis, and apoptosis. Activation of the coagulation pathway contributes to microvascular thrombosis.[47] In addition to factors related to the infection, AKI in critically ill patients with COVID-19 can be attributed to nonspecific factors including mechanical ventilation, hypoxia, hemodynamic instability, cardiac dysfunction, and the use of nephrotoxic agents.[47] Fig. 59.2 shows the pathogenesis of renal dysfunction in patients with COVID-19.

PATHOLOGY

Several studies investigating native kidney biopsy and autopsy specimens have consistently shown acute tubular necrosis (ATN) to be the predominant pathologic finding in individuals with AKI with COVID-19.[54–56]

Numerous glomerular diseases have also been documented in association with SARS-CoV-2 infection. The establishment of a definitive link between the development of glomerular disease and SARS-CoV-2 infection presents challenges because the role of this virus as either a primary cause or a contributing secondary hit in persons with a preexisting predisposition to glomerular pathology remains ambiguous. Several reports have documented the development of collapsing glomerulopathy, characterized by acute podocytopathy in the context of COVID-19, and the condition has been designated as COVID-19–associated nephropathy (COVAN).[57] Endothelial tubule-reticular inclusions are a characteristic sign of COVAN and are linked to high circulating levels of interferon. The high-risk *APOL1* genotype represents the main risk factor for the development of COVAN.[58] In addition, there have been reports of the development of new-onset podocytopathy (focal segmental glomerular sclerosis and minimal change diseases [MCD]) and vasculitis (pauciimmune crescentic GN and antiglomerular basement diseases) in association with COVID-19.[55,59,60] Cases of membranous nephropathy (MGN) have been reported, with the suggestion that the presence of phospholipase A2 receptor (PLA2R) in the respiratory tract may serve as a possible site for antigen presentation, inciting or enhancing anti-PLA2R responses.[55] Thrombotic microangiopathy (TMA) has also been reported in COVID-19, either as the predominant manifestation or alongside other findings such as ATN and collapsing glomerulopathy.[56,61] SARS-CoV-2 has been isolated from urine samples of patients with COVID-19.[62] Evidence for the presence of SARS-CoV-2 has been found in all kidney compartments examined using reverse transcription polymerase chain reaction (RT-PCR), in situ hybridization (ISH), and indirect immunofluorescence (IF) with confocal microscopy.[63,64]

Autopsies of kidney tissues obtained from patients with COVID-19 have revealed ATN and glomerulosclerosis.[54,55,60] Staining for S1 protein was found in renal parenchymal and tubular epithelial cells. Positive staining of NSP8, necessary for viral RNA synthesis and replication, suggests active viral replication in the kidney.[65] The detection of viral proteins by electron microscopy (EM) provides confirmatory evidence of SARS-CoV-2 infection in kidney tissue.[66,67] Single-cell RNA sequencing has revealed evidence of injury and activation of profibrotic signaling pathways in human-induced pluripotent stem cell–derived kidney organoids infected with SARS-CoV-2. Infection with SARS-CoV-2 also resulted in elevated expression of the collagen 1 protein in organoids, suggesting that SARS-CoV-2 can directly invade renal cells, leading to cellular damage and fibrosis.[68]

PROGNOSIS

AKI has been associated with an increased mortality in individuals hospitalized with COVID-19. In a study of 3993 COVID-19 patients hospitalized with COVID-19, the in-hospital mortality rate was 50% among individuals who had AKI, compared with 8% among those who did not develop AKI. Furthermore, the kidney function did not return to baseline in 35% of the AKI survivors. [28] In the STOP-COVID Cohort Study, which included 4221 ICU patients, more severe AKI was associated with increased mortality. Sixty-seven percent of the 876 patients who needed kidney replacement therapy (KRT) died, 11% had nonrecovery of kidney function, and 22% showed full kidney recovery at the time of discharge.[69] In a cohort analysis of patients with

Table 59.2 Characteristics Features of Various Infections with Associated Renal Syndromes

Infections	Causative Agent	Geographic Areas	Clinical Characteristics	Renal Syndromes	Renal Histopathology	Treatment
Bacterial Illnesses						
Leprosy	*Mycobacterium leprae*	Brazil, Africa, and southern Asia	Hypopigmented skin lesions on the skin, diminished or complete loss of sensation within localized areas of the skin, and paresthesia	Concentrating defects, AKI, CKD, Nephrotic syndrome, and GN	ATN, Proliferative GN, MPGN, MGN, amyloidosis, tubulointerstitial nephritis, granulomatous nephritis, and crescentic GN	Multidrug therapy with rifampicin, dapsone and clofazimine
Enteric fever	*Salmonella typhi*	African, Eastern Mediterranean, Southeast Asia, and Western Pacific Regions	Fever, abdominal pain, loss of appetite, and diarrhea	AKI and HUS	Acute nephritic syndrome, diffuse glomerulonephritis, AIN, immunoglobulin A (IgA)	Fluoroquinolones, azithromycin, and cephalosporins
Srub typhus	*Orientia tsutsugamush*	Southeast Asia, India, China, Japan, Indonesia, and northern Australia	Fever, eschar, malaise, lymphadenopathy, headache, gastrointestinal symptoms	AKI, proteinuria nephrotic syndrome, and hematuria	ATN, AIN, MGN, and mesangial hypercellularity with mesangial IGA deposition	Azithromycin and doxycycline
Leptospirosis	Spirochetes from the genus *Leptospira.* Rodents are the primary reservoirs responsible for transmission	South and Southeast Asia, Oceania, the Caribbean, sub-Saharan Africa, and Latin America	Fever, jaundice, headaches, bleeding diathesis, conjunctival hyperemia, gastrointestinal symptoms, and hepatosplenomegaly	Asymptomatic urinary abnormalities, AKI, electrolyte and tubular disorders, and CKD	ATN, AIN, mesangial proliferation, and vasculitis	Cephalosporins, doxycycline, and azithromycin
Viral Illnesses						
COVID-19	SARS-CoV-2	Pandemic	Cough, muscle pain, fever, dyspnea, sore throat, fatigue, headache, and anosmia	Proteinuria, hematuria, relapse of glomerular diseases, AKI, CKD, Vaccine-associated glomerular diseases	ATN, podocytopathy, COVAN, MGN, pauci-immune GN, and TMA	Per the severity (see text)
Dengue fever	Dengue virus. Primary vector is *Aedes aegypti* mosquito.	Africa, the Americas, the Eastern Mediterranean, Southeast Asia, and the Western Pacific	Headache, retro-orbital pain, fever, arthralgia, hematologic anomalies such as thrombocytopenia, increased hematocrit and leukopenia	Proteinuria, AKI, nephrotic, and HUS	Proliferative GN, MPGN, TMA, ATN, pigment nephropathy, and IgA mesangial deposits	Fluids and conservative management
Parasitic Illnesses						
Malaria	Caused by *Plasmodium.* Transmitted via female *Anopheles* mosquito	Sub-Saharan African region, Southeast Asia, Eastern Mediterranean, Western Pacific, and the Americas	Fever, accompanied by other symptoms such as chills and rigors, headaches, weakness, gastrointestinal discomfort, muscle pain, thrombocytopenia, and hepatosplenomegaly	AKI, hematuria, HUS, nephrotic syndrome, ESKD	ATN, AIN, collapsing nephropathy, MCD, ACN, immune-mediated proliferative GN, or TMA	Artesunate, artemether, and quinine

Table 59.2 Characteristics Features of Various Infections with Associated Renal Syndromes—cont'd

Infections	Causative Agent	Geographic Areas	Clinical Characteristics	Renal Syndromes	Renal Histopathology	Treatment
Leishmaniasis	Caused by obligate intracellular protozoa of the genus *Leishmania* transmitted by phlebotomine sandfly	Asia, Africa, the Americas, and Mediterranean region	Fever, malaise, weight loss, and splenomegaly	Proteinuria and hematuria, renal tubular dysfunction and AKI	AIN, proliferative and necrotizing GN, MPGN, and amyloidosis	Amphotericin B, pentavalent antimonial medicines, paromomycin and miltefosine
Filariasis	*Wuchereria bancrofti*, *Brugia malayi*, and *Brugia timori*. Main vector: *Culex* mosquito	Sub-Saharan Africa, Southeast Asia, the Indian subcontinent, the Pacific Islands, and specific regions in Latin America and the Caribbean	Acute adenolymphangitis, hydrocele, lymphedema, elephantiasis, chyluria, and tropical pulmonary eosinophilia, acute funiculitis, and epididymal-orchitis	Chyluria, microscopic hematuria, proteinuria, nephrotic syndrome, and AKI	Proliferative GN, MPGN, and renal amyloidosis	Diethylcarbamazine
Echinococcosis	Caused by various species of the tapeworm *Echinococcus.* Dogs are definitive hosts. Ungulates, including sheep and goats are intermediate hosts, and humans are aberrant intermediate hosts	Central Asia, western China, South America, eastern Africa, and Mediterranean countries	Asymptomatic, fever, dull flank pain, nausea, vomiting, and hydatiduria	Proteinuria and abdominal masses that resemble tumors	Case reports of MCD, MGN, MPGN, and amyloidosis	Surgical, albendazole
Fungal Infections						
Candidiasis	*Candida* species, such as *C. albicans C. glabrata, C. parapsilosis, C. kruse,* and *C. tropicalis*	Mostly manifests in individuals with compromised immune systems. Other predisposing factors include prolonged ICU stay, old age, presence of urinary drainage devices, diabetes, and urinary tract abnormalities	Fever, flank pain, oliguria, strangury, or passage of particulate materials and pneumaturia (suggest the presence of a fungal ball)	AKI and pyuria	Fungal granulomatous interstitial nephritis, microabscesses, renal infraction, and renal papillary necrosis	Fluconazole, amphotericin B
Mucormycosis	Most infections are caused by *Rhizopus* spp.	Mostly occurs in immunocompromised. Other risk factors include COVID-19 infection, trauma, burns, desferrioxamine therapy for iron/aluminum overload, and intravenous drug abuse	Fever, flank pain and tenderness, other systemic involvements such as cutaneous, rhino-orbitocerebral, pulmonary, and gastrointestinal can be there	Gross hematuria, pyuria, oliguria or AKI	Intrarenal abscesses. Vascular thrombosis can cause ischemia and necrosis of kidney tissue. Fungi may infiltrate the parenchyma, glomeruli, and tubules.	Debridement of involved tissue, amphotericin
Aspergillosis	Invasive cases are mostly attributed to species of *Aspergillus fumigatus* complex	Common in immunosuppressed patients. Can occur in patients with genitourinary instrumentation	Flank pain, fever, and systemic symptoms such as weight loss and exhaustion	Pyuria, hematuria, and AKI	Papillary necrosis, renal infarction, and renal abscesses	Voriconazole, amphotericin B, and surgical debridement

ACN, Acute cortical necrosis; *AIN,* acute interstitial nephritis; *AKI,* acute kidney injury; *ATN,* acute tubular necrosis; *CKD,* chronic kidney disease; *COVAN,* coronavirus 19–associated nephropathy; *ESKD,* end-stage kidney disease; *GN,* glomerulonephritis; *HUS,* hemolytic uremic syndrome; *ICU,* intensive care unit; *MCD,* minimal change disease; *MGN,* membranous glomerulonephritis; *MPGN,* membranoproliferative glomerulonephritis; *SARS-CoV-2,* severe acute respiratory syndrome coronavirus-2; *TMA,* thrombotic microangiopathy.

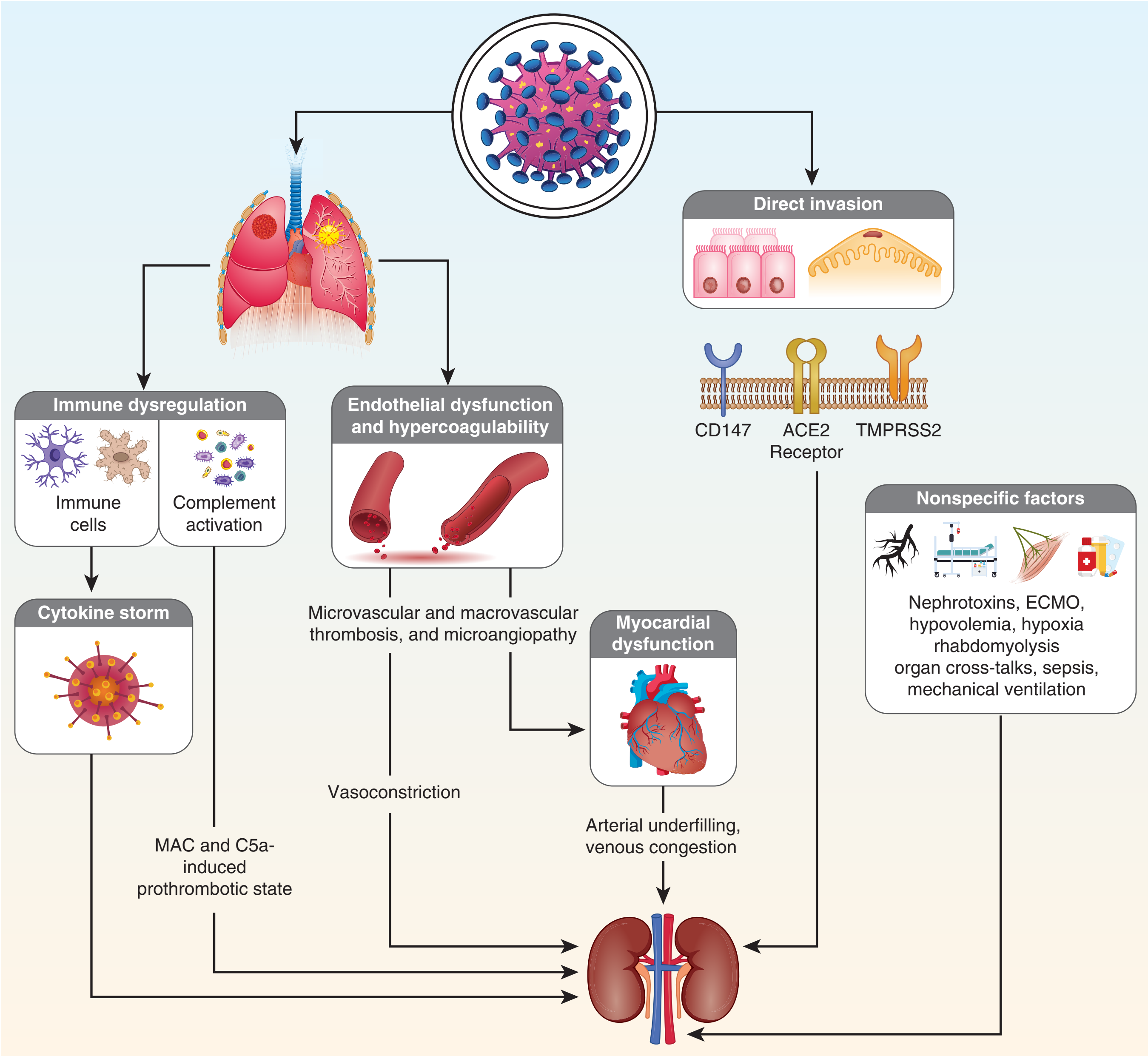

Fig. 59.2 Pathogenesis of kidney injury in patients with coronavirus 2019. *ECMO*, Extracorporeal membrane oxygenation; *MAC,* membrane attack complex.

in-hospital AKI (182 COVID-19–associated AKI and 1430 non–COVID-19 AKI), COVID-19–related AKI was associated with a greater rate of estimated glomerular filtration rate (eGFR) decline after discharge, regardless of underlying comorbidities or AKI severity. The fully adjusted model controlling for comorbidities, peak creatinine level, and in-hospital dialysis requirement showed a greater reduction in eGFR in COVID-19–associated AKI (–14.0; 95% CI, –25.1 to –2.9 mL/min/1.73 m^2/year; $P = 0.01$). This eGFR trajectory emphasizes the need for long-term monitoring of renal function in patients with COVID-19–associated AKI.[70]

After controlling for demographics and comorbidities, population-based studies show a higher mortality rate in dialysis patients with COVID-19.[71,72] The age-matched relative risk of death from COVID-19 for an in-center hemodialysis patient compared with the general population ranged from 432 for a 20- to 39-year-old patient to around 10 for a patient aged >80 years.[73]

COVID-19 AND KIDNEY TRANSPLANTATION

The COVID-19 pandemic intensified the challenges associated with organ transplantation. Transplant centers globally experienced a decrease in the number of kidney donations from both deceased and living donors. KTRs with COVID-19 had increased fatality rates, likely attributable to compromised immune responses resulting from prolonged administration of immunosuppressive medications, as well as the high prevalence of comorbid conditions such as diabetes and cardiovascular diseases.[74,75] A retrospective analysis of graft-related outcomes in the United Kingdom revealed that transplantations performed before and during the COVID-19 pandemic exhibited similar rates of primary nonfunction, delayed graft function, acute rejection, reoperation, length of hospital stay, and graft survival.[76] A study comparing 56 patients waitlisted for a kidney transplant and 80 KTRs diagnosed with COVID-19 showed that despite similar demographic profile and comorbidity burden, waitlisted patients had higher rates of hospitalization and an elevated mortality risk.[77] Whether immunosuppression should be reduced in KTRs with COVID-19 is not clear. While immunosuppression reduction has been a common practice, some studies have found that reducing immunosuppression may increase the risk of graft rejection.[78,79] Therefore the decision to reduce immunosuppression, typically involving antimetabolites or calcineurin inhibitors, should be individualized, considering the balance between disease severity and risk of rejection.[76,79,80]

MANAGEMENT

Management of COVID-19 is primarily dictated by patient age, underlying health conditions, and disease severity. In addition to supportive measures, the range of therapeutic approaches includes antiinflammatory agents, anticoagulants, monoclonal antibodies, and antiviral therapies, individually or in combination. Despite being an important risk factor for severe and fatal COVID-19, clinical trials of drugs for COVID-19 have systematically excluded individuals with impaired kidney function. As a result, the appropriate dosage, frequency, and safety of these pharmaceuticals for individuals with CKD are not clear.

Several guidelines have been developed for the pharmacotherapy of COVID-19. All recommend the use of dexamethasone for severe COVID-19.[81] The antiviral remdesivir, an RNA-dependent RNA polymerase inhibitor, is recommended for hospitalized individuals. Sulfobutylether-β-cyclodextrin, which serves as a carrier for intravenous administration of remdesivir, has the potential to accumulate within the renal tubules and cause nephrotoxicity.[82] Remdesivir has, however, been used safely in COVID-19 patients with AKI and CKD.[83–85] A study on hospitalized COVID-19 patients with eGFR from 15 to 60 mL/min/1.73 m^2 found no statistically significant differences in peak creatinine levels during hospitalization, creatinine doubling, or KRT initiation between remdesivir-treated patients and matched untreated historical comparators.[84] The U.S. Food and Drug Administration has approved remdesivir for treating COVID-19 in patients with advanced kidney disease including those on dialysis.[86]

The combination of remdesivir and baricitinib, a Janus kinase inhibitor, reduces the recovery time in patients with COVID-19. This effect is particularly significant for individuals receiving high-flow oxygen or noninvasive ventilation.[87] The use of nirmatrelvir-ritonavir was associated with a decrease in all-cause hospitalization and all-cause mortality in patients with CKD and COVID-19.[88,89] Vilobelimab, an anti-C5a monoclonal antibody that specifically targets an inflammatory pathway, also received Food and Drug Administration authorization for use in hospitalized COVID-19 patients who have initiated invasive mechanical ventilation or extracorporeal membrane oxygenation within a 48-hour duration.[90]

VACCINES

Various vaccines for COVID-19 have been developed, such as the mRNA vaccines BNT162b2 (Pfizer/BioNTech) and mRNA-1273 (Moderna), the viral vector vaccine ChAdOx-1 nCoV-19 (University of Oxford/AstraZeneca), the recombinant spike protein (S-protein) subunit vaccine NVX-CoV2373 (Novavax), and the inactivated whole-virus SARS-CoV-2 vaccine.[91] Given the increased risk of acquiring the SARS-CoV-2 infection and the case fatality rate from COVID-19, patients with CKD, especially those on dialysis, have been prioritized for vaccination in many countries.[92]

In general, the seroconversion rate following the COVID-19 vaccine in dialysis patients is lower than that in the general population.[93,94] The Dutch REnal Patients COVID-19 VACcination (RECOVAC) Consortium evaluated the immunogenicity of the mRNA-1273 COVID-19 vaccine in CKD stages G4–5, KTRs, and controls. Transplant recipients had a lower rate of seroconversion compared with the control group (56.9% vs. 100%, $P<0.001$). The study showed a high seroconversion rate among those with CKD G4-5 (100%) and those undergoing dialysis (99.4%). However, the average antibody concentrations in both the CKD G4-5 cohort and dialysis cohort were comparatively lower than those in the control group.[95] In contrast, some studies have shown reduced seroresponse rates in patients undergoing dialysis after receiving two doses of the COVID-19 vaccine, as compared with healthy controls (≈80%).[93,94] Furthermore, the findings of the Dia-Vacc study indicate that over time, both dialysis patients and KTRs are at risk of a decline in IgG and receptor-binding domain-IgG (RBD-IgG) antibodies. Six months after vaccination, 98% of the control group continued to show antibodies, whereas the rate had dropped in the dialysis patients and KTR groups to 91% and 87%, respectively ($P = 0.005$). The peak antibody titers are lower in KTRs, with an accelerated drop thereafter.[96]

Suboptimal immunogenicity, waning antibody levels, breakthrough cases in fully vaccinated individuals, and ambiguous efficacy against COVID-19 have prompted the implementation of supplementary booster doses of the COVID-19 vaccine for patients undergoing dialysis and KTRs.[97,98] Housset and colleagues[99] evaluated the effects of a fourth dose of mRNA vaccination administered after a median duration of 7.6 months following the third dose in dialysis patients. They showed a significant antibody response and raised antispike antibody titers.[99] There is no discernible disparity in the safety profile of COVID-19 vaccination between individuals undergoing dialysis and the general population.[100]

In CKD and dialysis patients, the incidence of mortality risk is significantly lower in those who are vaccinated than nonvaccinated.[31,101,102] In a retrospective analysis including adult patients undergoing chronic dialysis, after adjusting for age, sex, and Charleston comorbidity index, vaccinated patients had a significantly decreased composite risk of mortality or hospitalization (odds ratio 0.24, 95% CI 0.15–0.40).[99]

The effectiveness of the COVID-19 vaccination in KTRs is suboptimal.[96] A meta-analysis showed that KTRs had a lower seropositivity rate (26.1%) after two COVID-19 vaccination doses than patients on hemodialysis (84.3%) or peritoneal dialysis (92.4%).[103] Vaccines had similar side effects in KTRs as in phase III studies of the general population. No instances of the development of donor-specific antibodies following vaccination were reported.

There have been reported instances of new-onset and relapsed kidney diseases following the administration of COVID-19 vaccines. The nephropathies reported post COVID-19 vaccination include MCD, immunoglobulin A nephropathy (IgAN), MGN, antineutrophilic cytoplasmic antibody (ANCA)-associated vasculitis, acute interstitial nephritis (AIN), and TMA.[104–108] The reported incidence of cases is small compared with the vast number of vaccine doses administered, and the advantages of COVID-19 immunization significantly outweigh any possible risks.

DENGUE

Dengue is caused by the positive-sense single-stranded RNA virus of the genus *Flavivirus*. The virus has four serotypes, each characterized by specific antigenic properties and genotypic variations. The principal vectors responsible for virus transmission are the mosquito species *Aedes aegypti* and *Aedes albopictus*.

EPIDEMIOLOGY AND CLINICAL FEATURES

Annually, around 400 million cases of dengue are reported globally, resulting in roughly 22,000 fatalities.[109] The disease is endemic in more than 100 countries including regions of Africa, the Americas, the Eastern Mediterranean, Southeast Asia, and the Western Pacific. According to the World Health Organization (WHO), the incidence of dengue has increased from approximately 505,000 in 2000 to 5.2 million in 2019.[110] Disease transmission is affected by the ambient temperature and the virus genotype.[111] Ongoing global warming is creating an environment hospitable to the spread of both the Aedes mosquito and the virus, leading to the spread of the disease in the subtropical parts of the world, such as Europe, where it did not exist until recently. The number of imported dengue cases in France went up from 217 in 2022 to 1414 in 2023, and new-onset local transmission of the disease has been documented in Italy, Spain, and France.[112]

Disease presentation ranges from mild fever to severe, life-threatening multiorgan involvement. In 1977 the WHO classified dengue infection into asymptomatic infection, dengue fever (DF), dengue hemorrhagic fever (DHF), and dengue shock syndrome (DSS).[113] In 2009, this classification scheme was revised to dengue without warning signs, dengue with warning signs, and severe dengue. Patients with kidney failure on dialysis are at risk of developing severe disease.[114]

The ratio of asymptomatic to apparent infections is about 15:1.[115] The disease has three phases: febrile, critical, and recovery. The main features include fever, myalgia, headache, arthralgia, retroorbital pain, body rash, lymphadenopathy, and hematologic anomalies including thrombocytopenia, increased hematocrit, and leukopenia. DHF is distinguished by the presence of increased vascular permeability and profound vascular leakage. DSS is a severe complication characterized by hypovolemia and compromised peripheral perfusion, leading to tissue damage and multiorgan failure. Individuals are at higher risk for severe dengue infection when they experience sequential dengue infections with two different strains of dengue virus.

Kidney manifestations of DF include proteinuria, nephrotic syndrome, and AKI.[116–118] Cases of hemolytic uremic syndrome (HUS) characterized by AKI and clinical and histologic signs of TMA have been reported in DF.[119] The prevalence of dengue-associated AKI varies across studies, with reported rates ranging from 0.9% to 70%.[120–122] In a large retrospective cohort involving 1484 individuals, 4.8% of patients had AKI, and 14.1% of those with AKI underwent dialysis.[123]

The diagnosis of dengue can be made by detecting viral genetic material by RT-PCR or viral antigen nonstructural protein 1 (NS1). In primary infections, the sensitivity of NS1 detection exceeds 90%.[124] The diagnosis can also be confirmed by demonstrating a rising antibody titer (IgM and/or IgG).

PATHOGENESIS

Dengue virus can directly or indirectly damage the kidney through shock, renal hypoperfusion, or immunologic dysfunction.[23] Other possible mechanisms include hemolysis and rhabdomyolysis. DHF/DSS represents the development of an aberrant and exacerbated immune response by the host.[125] Initial activation of $CD4^+$ cells initiates both Th1-mediated and Th2-mediated responses, leading to the release of multiple cytokines, namely interleukin-2 (IL-2), interferon-γ (IFN-γ), and tumor necrosis factor-α (TNF-α), which can cause tissue injury. Viral antigens have been documented in glomerular and tubular tissues, suggesting direct tissue damage.[126,127]

PATHOLOGY

Renal histopathology in dengue patients has shown a range of lesions including proliferative GN, membranoproliferative glomerulonephritis (MPGN), TMA, ATN, pigment nephropathy, and IgA mesangial deposits.[117–119,127–129] IF shows the presence of glomerular IgG and/or IgM and C3 deposits, suggesting immune-mediated injury. Microtubuloreticular structures have been identified on EM, indicating a possible viral infection.[119]

MANAGEMENT

Treatment of DF is largely supportive. In the early stages of the disease, care involves monitoring vital parameters such as blood pressure, hematocrit levels, platelet count, urine output, and consciousness. Prompt restoration of the circulation volume by using crystalloid solutions is crucial. Platelet transfusions may be warranted in patients with severe thrombocytopenia ($<10{,}000/mm^3$) and active bleeding.

AKI should be managed in the usual way with supportive treatment including KRT as needed. Recovery from uncomplicated dengue usually takes 2 weeks. In one study, the mortality rate among patients with AKI was 12.7%,[123] whereas in other studies, the case fatality rate among individuals diagnosed with DF and AKI was around 60%.[120,121,126] Malhi and colleagues[130] showed that among survivors of patients with dengue-associated AKI, 23% showed persistently low eGFR (<60 mL/min/1.73 m^2) 3 months after discharge.

Two dengue virus vaccines (Dengvaxia and Qdenga) have been shown to protect people with laboratory-confirmed prior dengue virus infection from dengue infection by 60% to 80% and to prevent severe dengue that leads to hospitalization by 70% to 90%. These vaccines have been recommended by the Centers for Disease Control and

Prevention Advisory Committee on Immunization Practices and WHO. The Dengvaxia vaccine is approved for use in individuals aged 9 through 45 years, whereas the WHO recommends Qdenga for children aged 6 to 16 years in high dengue-burdened areas.[131] These vaccines, however, raise the risk of severe dengue infection in dengue-naïve children and hence are not recommended. In a recent randomized controlled trial, administration of the Butantan-Dengue vaccine was effective in preventing symptomatic infections of DENV-1 and DENV-2, irrespective of dengue serostatus, over a 2-year follow-up period.[132] A systematic review also showed that dengue vaccines could confer protection against severe dengue in pediatric populations.[133]

An innovative biocontrol strategy for dengue control is infecting the *A. aegypti* mosquitoes with the gram-negative bacterium *Wolbachia*.[134] Upon release in endemic areas, *Wolbachia*-infected mosquitoes mate with uninfected mosquitoes, reducing their reproductive capabilities, lifespan, and vector competence through several mechanisms. This approach might allow vector control without posing risks to natural ecosystems or requiring genetic modification of the mosquitoes.

MYCOBACTERIAL AND RICKETTSIAL INFECTIONS

TUBERCULOSIS

Tuberculosis (TB), one of the most prevalent infectious diseases worldwide, is caused by bacilli considered as variants of a single species, the *Mycobacterium tuberculosis* complex. Other mycobacteria, found mainly around water sources, are called environmental mycobacteria and can also cause disease.

EPIDEMIOLOGY AND CLINICAL FEATURES

The global prevalence of *M. tuberculosis* (Mtb) infection is estimated at approximately 2 billion. However, only a minority of infected people develop TB. Annually, the number of people becoming ill with TB is estimated at 10 million, with 1.3 million deaths.[135] Most of these individuals reside in low- and middle-income countries, with approximately 50% in Bangladesh, China, India, Indonesia, Nigeria, Pakistan, the Philippines, and South Africa.[136] Tuberculosis is the main contributor to infectious disease mortality on a global scale.

Risk factors for the development of TB include malnutrition, human immunodeficiency virus (HIV) infection, diabetes mellitus, chronic kidney and liver disease, homelessness, poor housing conditions, and the use of immunosuppressive medicines. Patients with kidney failure including those on dialysis and KTRs have a 10- to 50-fold greater TB rate than the general population.[137] The primary mode of disease transmission is interpersonal, through inhalation of droplet aerosols containing bacilli. Additional mechanisms are prenatal transmission, accidental inoculation, sexual transmission, and therapeutic instillation.[138–140]

After primary infection, Mtb bacilli undergo local multiplication in tissues and trigger a diverse range of immunologic responses. These reactions can lead to either eradication of the bacteria or their containment through the formation of granulomas, also known as the "primary Ghon focus." Primary TB lesions are commonly observed in the lungs, tonsils, or gastrointestinal tract, although all organs can potentially serve as a site of infection. The bacilli travel through the lymphatics to the nearby lymph nodes. The infection remains contained because of slow multiplication, intracellular localization inside macrophages, and development of acquired immune responses. Any immunosuppressed status, such as CKD, that diminishes the activity of cell-mediated immune reactions facilitates the multiplication of tubercle bacilli.[141] This persistent and dynamic interaction between the organism and host immune response determines the development of sequelae-like caseous necrosis, abscess, ulceration, fistulae, fibrosis, or calcification if the host's immune system is unsuccessful in eliminating MTB.

Urogenital tuberculosis (UG-TB) encompasses the involvement of several organs within the genitourinary system including the kidneys, ureters, bladder, prostate, urethra, penis, scrotum, testicles, epididymis, vas deferens, ovaries, fallopian tubes, uterus, cervix, and vulva. Two studies from the United Kingdom reported the frequency of UG-TB to be 27% in 1989 and 1.8% in 2019. In addition to the natural reduction in prevalence, varying notification practices could have accounted for this difference. Only a small percentage of people diagnosed with kidney TB have concurrent pulmonary TB, with approximately 10% of cases presenting with active pulmonary TB.[142] About one-third of patients exhibit an abnormal chest radiograph.[142,143] In India, genitourinary TB comprises 20% of all extrapulmonary TB and is the most common extrapulmonary system to be affected by this disease.[144]

For reasons that are not clear, UG-TB is encountered twice as frequently in men as women.[145] The disease is seen more frequently among ethnic minorities in developed countries. In Western countries, UG-TB presents in the fifth to seventh decades, whereas no age predilection is noted in developing countries. There are no specific presenting features, often causing the disease to be overlooked until the late stages. One of the classical presentations is sterile pyuria in someone with dysuria and/or frequency. The typical symptoms of TB, such as fever, night sweats, nocturia, and weight loss, are not common but strengthen clinical suspicion when present. Rare symptoms include back, flank, or abdominal pain and macroscopic hematuria. Genital involvement can present with epididymo-orchitis or prostatitis, manifesting with pelvic discomfort, dysuria, hematospermia, sexual dysfunction, hesitancy, and difficulty in urination. There have been case reports of patients presenting with AKI and found to have granulomatous interstitial nephritis on kidney biopsy. Resolution of AKI following antitubercular therapy strengthens the etiologic association.

DIAGNOSIS

UG-TB may go overlooked due to its nonspecific symptoms, long-lasting and ambiguous clinical presentations, and a lack of knowledge among health care professionals. Diagnosis requires a high index of suspicion, with confirmation by a bacteriologic examination or tissue biopsy. Demonstration of TB bacilli in the urine requires repeated (three to six times) testing on successive days of early-morning urine samples for acid-fast bacilli (AFB) by Ziehl Nielsen staining, mycobacterial culture, or PCR.[146] The AFB smear microscopy technique requires a minimum concentration of 5×10^3 bacilli per milliliter of the specimen.[147,148] Light-emitting diode-based fluorescence microscopy has replaced traditional Ziehl Nielsen staining because it provides a faster diagnosis. Further, the demonstration of AFB is not always diagnostic since the urinary tract may be colonized by environmental mycobacteria, especially in swimmers. The culture of clinical specimens for Mtb is widely regarded as the most reliable diagnostic approach for identifying active TB, with a specificity of 100% and a sensitivity of 65%.[146,147,149] Historically, the conventional approach involved employing the solid Lowenstein-Jensen culture medium, which necessitated a time frame of 6 to 8 weeks. This method has been substituted by the automated liquid mycobacteria growth indicator tube (MGIT) culture, which employs the BACTEC MGIT 960 System (Becton Dickinson, Franklin Lakes, New Jersey, USA) and can confirm the growth of microorganisms in approximately 2 weeks.[148] The GeneXpert MTB/RIF assay is a cost-effective and expeditious diagnostic tool. It enables the simultaneous detection of Mtb and rifampicin resistance.[149] GeneXpert demonstrates a superior performance compared with AFB smear and culture in detecting M tuberculosis in urine samples.[150] The TB-LAM (lipoarabinomannan) urine test is recommended for the diagnosis of HIV-associated TB because patients with advanced immunosuppression are at an increased risk of disseminated Mtb infection, which results in renal involvement and the release of heat-stable Mtb LAM glycolipid into the urine.[151]

Radiologic tests (intravenous urography, computed tomography scans, or ultrasound) that show concomitant upper and lower urinary tract involvement strongly support a diagnosis of UG-TB. Classical radiologic features include mucosal thickening, cavities, strictures, calcification, and many other manifestations (Fig. 59.3).[152] Imaging facilitates the precise guidance of procedures such as abscess aspiration or tissue biopsy, allowing for microbiologic and molecular evaluation. Those with negative bacteriologic, nucleic acid, or radiographic findings may need a tissue biopsy of the urinary tract or the kidney showing caseating granulomata with or without AFB. A rare manifestation is hypercalcemia due to the aberrant extrarenal synthesis of 1,25-dihydroxyvitamin D3 by activated macrophages within the granulomatous tissue.[153]

PATHOGENESIS

Tubercle bacilli can colonize the kidneys through hematogenous or lymphatic spread after the original infection from the lungs or gut. The initial seeding is usually within the cortex, near the glomeruli or the peritubular capillary bed. When the bacilli pass into the nephrons, they become entrapped within the loop of Henle, producing new foci of infection. The resulting granulomatous inflammation and its progression contribute to the development of chronic tubulointerstitial nephritis, papillary necrosis, fibrosis, and extensive caseous destruction of the renal parenchyma. Multiple microscopic foci of necrosis can coalesce, giving rise to macroscopic lesions that involve the renal papilla.[147] Destructive inflammation, cavities, and vascular insufficiency cause papillary necrosis.[154] Tubercular pyelonephritis may progress to pyonephrosis with increasing fibrosis and scarring of the renal pelvis with dilation of the calyces.[155] These processes unfold over the years.

Renal parenchymal involvement in TB (Fig. 59.4) has been classified into four stages. Stage 1 is a nondestructive kidney parenchymal TB, which is evident only histologically. Minimal destruction associated with TB papillitis characterizes stage 2. More destruction leads to the formation of cavities (stage 3), and the final stage 4 is the extensive and highly destructive polycavernous kidney TB.[156] Dystrophic parenchymal calcification is common and may be seen in stages 2 to 4. Unabated progression can lead to the development of "putty kidney" characterized by the presence of sacs containing caseous, necrotic material, and dystrophic calcification, often associated with autonephrectomy.[157]

PATHOLOGY

TB can involve all compartments of the kidneys and urinary tract. The most common presentation is chronic tubulointerstitial nephritis with granulomas.[158] Chronic TB

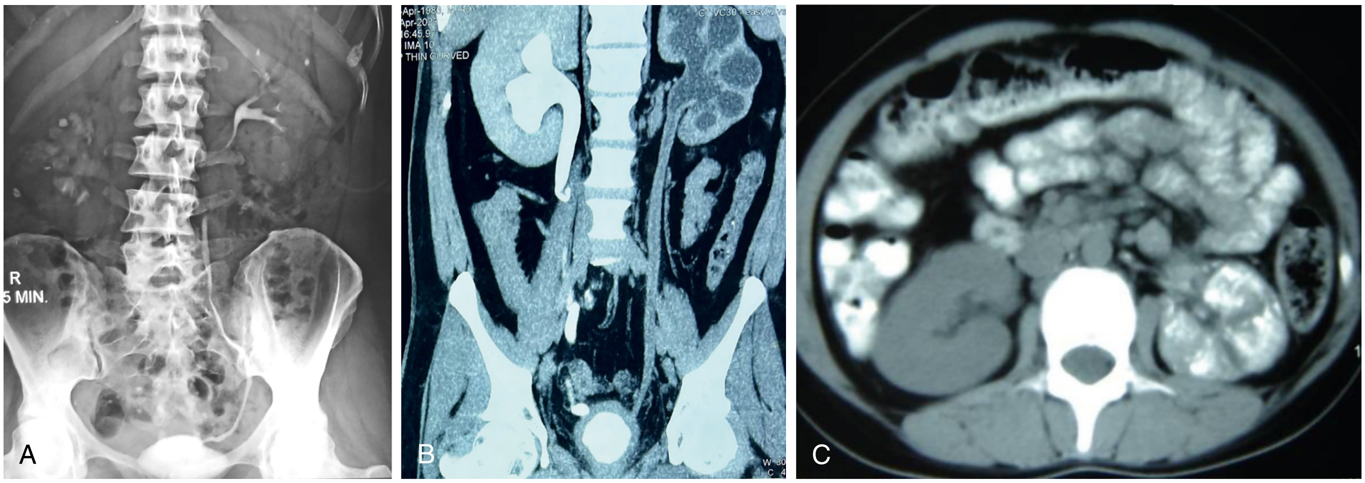

Fig. 59.3 TB radiology: radiologic abnormalities in genitourinary tuberculosis. (A) An intravenous urogram showing infundibular stenosis, pelvic scarring, irregular dilatations of calyces, and cavitations of the right kidney. (B) Computed tomography (CT) urography showing destruction of parenchyma of left kidney (autonephrectomy), with thickened pelvicalyceal system and ureter. Right kidney is obstructed due to poorly compliant, fibrosed, and thickened bladder wall with reduced capacity. (C) Plain CT scan showing lobar calcifications in the left kidney. (Image courtesy Dr. Anupam Lal and Dr. Shrawan K. Singh.)

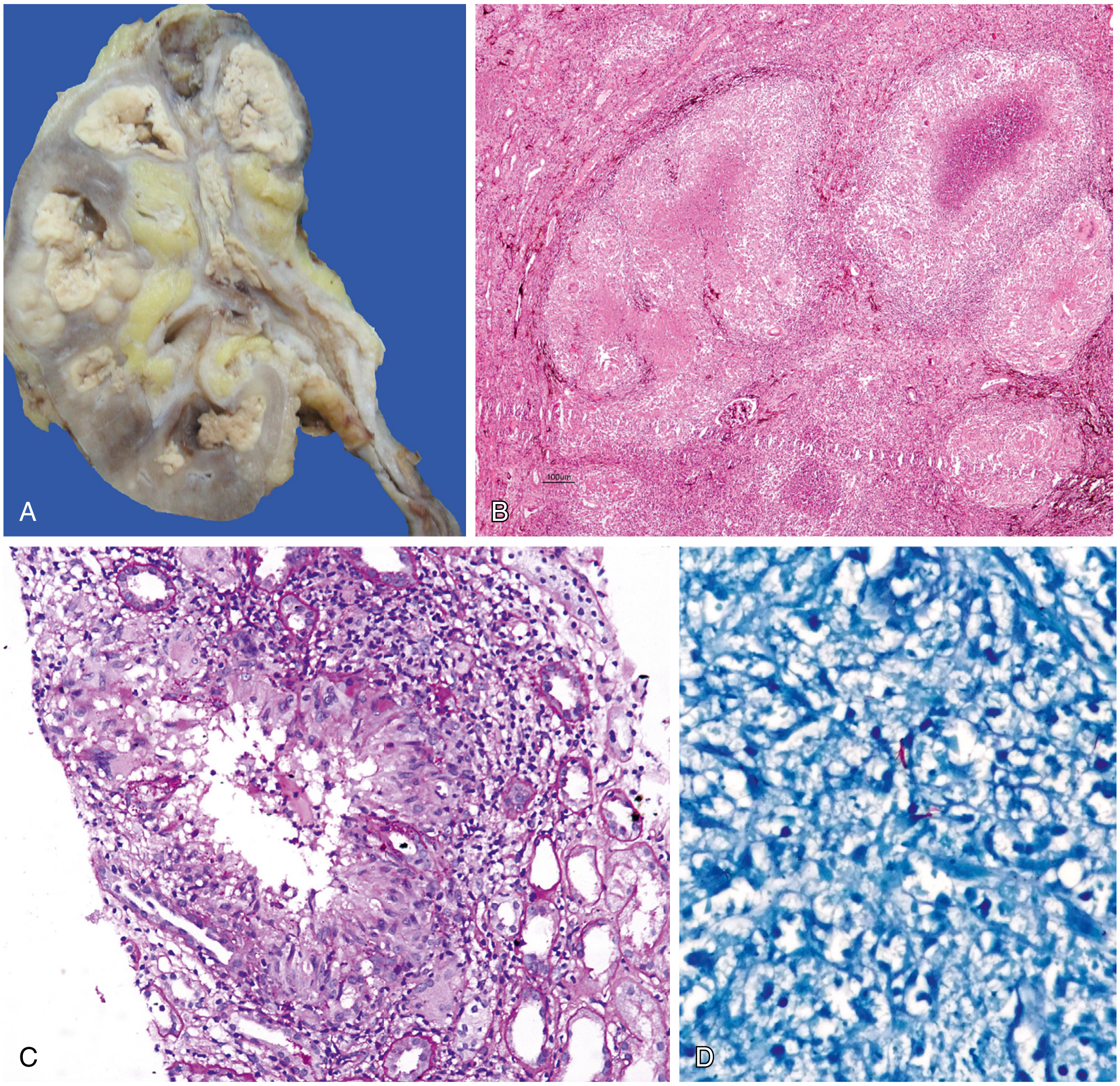

Fig. 59.4 (A) Gross image of end-stage kidney disease secondary to multiple cavities in the calyces, filled with gray-white friable pultaceous material. The ureteric wall is thickened, forming megaureter; (B) Scanner and low magnification view of kidney tissue showing numerous granuloma with central necrosis, admixed with Langhan multinucleate giant cells (PAS stain); (C) Core biopsy of the kidney shows epithelioid cell granuloma with central breakdown and Langhan giant cells (PAS, 20×); (D) Acid-fast bacillus in renal tissue (Ziehl Nielsen stain, 100×). (Image courtesy Dr. Mahesha Vankalakunti.)

may lead to the development of secondary amyloidosis.[159] The relationship between TB and other types of GN remains inconclusive, with several case reports of focal proliferative GN with the presence of immune system deposits, MGN, mesangiocapillary glomerulonephritis, and crescentic GN.[159–163]

Inflammation from renal TB can extend into the perirenal and pararenal regions including the psoas sheath, resulting in the formation of cold abscesses, sinus tracts, and fistulae. mTB bacilli from the kidney spread into the ureters to cause inflammation, edema, granulomatous ulceration, and fibrosis. These pathogenic processes cause irregular ureteric strictures, segmental dilatation, obstruction, and reflux.[164] Chronic inflammation and ureteric strictures lead to hydroureteronephrosis. An alternating pattern of nonconfluent dilatations and strictures may resemble a corkscrew or string of beads. The ureter may become shortened and stiff, known as a “pipe-stem ureter,” lacking peristaltic movement.[165]

Other features include cystitis and fibrosis, stenosis, or strictures at the ureterovesical junction (giving the orifice a golf-hole appearance) and hydroureteronephrosis.[166] Chronic bladder wall and detrusor muscle inflammation can cause thimble bladders (small capacity with an elevated base) due to thickening of the bladder wall with trabeculation and calcification. Vesicovaginal, vesicocolic, or enterovesical fistulae and bladder perforation are rare consequences of bladder TB. The retrograde spread of TB from the prostate or testes to the bladder has also been described.[154]

Bilateral disease can lead to gradual deterioration in kidney function, ultimately leading to kidney failure. In a retrospective analysis of 56 patients with UG-TB, 20% and 7% of patients developed CKD and ESKD, respectively.[167] AKI and old age were independent risk factors for CKD.

MANAGEMENT

Treatment of drug-sensitive tuberculosis involves a 2-month intensive phase of chemotherapy consisting of daily administration of first-line tuberculosis medications, rifampicin, isoniazid, pyrazinamide, and ethambutol along with pyridoxine to prevent isoniazid-induced neurotoxicity.[168] This is followed by a 4- to 6-month continuation phase in which two drugs, rifampicin and isoniazid, are given. Drug-resistant TB requires 18 to 24 months of treatment with potentially toxic second- or third-line agents.

No dosage adjustment is needed for isoniazid and rifampicin in individuals with renal impairment.[169] The dose of pyrazinamide and ethambutol is halved in patients with a glomerular filtration rate (GFR) below 10 mL/min/1.73 m^2, including those undergoing dialysis. Drugs can be given in full doses on alternate days or daily at 50% of the prescribed dose. It is advisable to use caution while administering streptomycin and ethambutol to patients with CKD due to the potential risks of ototoxicity and ocular toxicity, respectively.

Surgical interventions may be needed in certain circumstances including drainage of the obstructed pelvicaliceal system or abscesses, nephrectomy for nonfunctioning kidneys, ureteric reconstruction procedures, and reconstructive surgery of the bladder to improve functional bladder capacity reduction.[170] Definitive surgery is usually delayed until after the active disease has been treated with chemotherapy. Stenting or percutaneous nephrostomy may be indicated for patients with ureteral strictures, especially if hydronephrosis is also present. In a study of 77 patients, early ureteral stenting or percutaneous nephrostomy was associated with a lower nephrectomy rate (73 vs. 34 percent).[171] Nephrectomy may be considered in cases of extensive unilateral disease. Urinary tract lesions may progress despite effective treatment, and new lesions may appear due to inflammation and/or fibrosis.

LEPROSY

Leprosy is a chronic infectious disease caused by the bacterium *Mycobacterium leprae* (seen worldwide) and *M. lepromatosis* (reported from the Americas).[172] It occurs in more than 120 countries, with Brazil, India, and Indonesia contributing to the majority of the cases, followed by other South Asian and sub-Saharan African nations.[173]

CLINICAL CHARACTERISTICS

Leprosy is transmitted by prolonged, close contact with individuals with untreated leprosy, usually through respiratory droplets. The disease has a long incubation period of several years. It presents with predominant cutaneous (hypopigmented or erythematous skin lesions or skin thickening) or neurologic involvement (paresthesia, hypoesthesia, or thickened nerves).[172] The diagnosis is made by demonstration of typical histologic findings on full-thickness biopsy of the involved skin and nerves or a PCR for identification of *M. leprae* deoxyribonucleic acid (DNA). The disease spectrum is also affected by the host immune response—with fewer lesions and bacilli when the response is robust (paucibacillary or tuberculoid disease) and a greater number of lesions and higher bacillary load on biopsy in the case of a weaker immune response (multibacillary or lepromatous disease).

A precise estimate of the frequency of renal involvement cannot be made, with the presentation ranging from asymptomatic tubular function abnormalities to varying degrees of proteinuria and, less commonly, kidney failure.[174] In one study, abnormalities in urinary concentrating ability were found in 83% and 85% of patients with paucibacillary and multibacillary leprosy, respectively.[175] In another retrospective analysis from Brazil evaluating 923 patients, 4.8% exhibited proteinuria, 6.8% had hematuria, 10.4% showed leukocyturia, and 3.9% showed elevated serum creatinine. Most individuals had multibacillary leprosy. Hematuria is more prevalent among patients with multibacillary lepromatous leprosy, particularly during episodes of lepra reaction (an inflammatory response due to immune system activation, affecting 30% to 50% of patients with leprosy). A reduced GFR is associated with reaction episodes, multibacillary infections, and advanced age. Nephrotic syndrome is seen in those with amyloidosis, more frequently in those with lepromatous disease.[18,176,177] Multidrug therapy (MDT) has reduced the frequency of kidney disease. In one recent study, 45.5% of patients exhibited elevated serum creatinine at diagnosis, which decreased to 18% and 9% after 3 and 8 months of MDT.[178] AIN secondary to dapsone or rifampicin has been reported rarely.[179,180]

PATHOGENESIS

The precise mechanism underlying kidney injury in leprosy remains incompletely elucidated, with immune complex–mediated injury resulting from the interactions between antigens originating from *M. leprae* and elevated levels of humoral antibodies targeting these antigens being the primary hypothesis.[181] Abnormal cell–mediated immunity has also been demonstrated. Serologic investigations have indicated a decrease in complement hemolytic (CH50) activity and diminished levels of C4 in individuals with lepromatous leprosy, suggesting the activation of the classical or lectin pathway.[182–184]

PATHOLOGY

A wide range of histopathologic lesions have been observed. In an autopsy study of 199 patients from Brazil, 72% displayed renal lesions.[185] The lesions included ATN, AIN, MGN proliferative GN, crescentic GN, MPGN, and amyloidosis. Epithelioid granulomas and lepra bacilli were also encountered. IF investigations reveal granular deposits of IgG within the mesangium and along glomerular capillary loops with variable deposits of C3, IgA, IgM, and fibrin.[176,185] EM shows electron-dense deposits in the mesangial, subendothelial, and subepithelial areas.[185,186]

MANAGEMENT

Treatment entails standard multidrug therapy with dapsone, rifampicin, and clofazimine for 6 to 12 months. No specific treatment is required for renal lesions.[187] A short course of corticosteroids may be administered to those with AIN.

SCRUB TYPHUS

Scrub typhus, caused by the rickettsial bacterium *Orientia tsutsugamushi,* is a growing health concern in South and Southeast Asia, China, Japan, and northern Australia, all part of the "tsutsugamushi triangle."[188] Globally, more than a billion people are at risk of getting the infection, with over

1 million cases reported annually. This is likely to be an underestimate, given the absence of a surveillance system and large number of paucisymptomatic cases.

Humans get the infection through the bite of trombiculid mite larvae, also known as "chiggers." The clinical manifestations typically emerge following an incubation period ranging from 7 to 10 days, commonly presenting as fever, headache, myalgia, lymphadenopathy, and gastrointestinal symptoms (nausea, vomiting, diarrhea). Infected individuals can have a typical eschar characterized by the formation of a black ulcer with central necrosis after a primary papular lesion.[189] The eschar may be easily overlooked, especially in dark-skinned individuals.

Kidney involvement is frequently reported in scrub typhus. Urinary abnormalities are found in 40% to 60% of cases, with albuminuria (55%) and microscopic hematuria (16%) being the most common.[190,191] AKI develops in 20% to 50% of the cases[190–193] and may require dialysis in 3% to 10%.[191,193] A retrospective analysis of 510 consecutive scrub typhus patients in South Korea revealed an AKI incidence rate of 35.9%.[192] CKD, advanced age, lower blood albumin levels, and delayed hospital presentation were all reported as significant risk factors for AKI. In recent years, scrub typhus has emerged as the dominant cause of AKI in the setting of an acute febrile illness in South Asia.[194]

PATHOGENESIS

Kidney involvement in scrub typhus is multifactorial—attributable to systemic inflammatory response syndrome, endothelial dysfunction, volume depletion, disseminated intravascular coagulation (DIC), rhabdomyolysis, and direct *O. tsutsugamushi* infiltration into the renal parenchyma, as demonstrated by the presence of the organisms within the renal tubules.[195–197] Contributory factors include the need for intensive care, thrombocytopenia, pneumonia, shock, and acute respiratory distress syndrome (ARDS).

Respiratory distress, encephalitis, AKI, myocarditis, septic shock, and multiple organ dysfunction syndrome (MODS) are the leading causes of death among patients with severe disease.[191,193]

PATHOLOGY

Descriptions of renal pathology in scrub typhus are mostly in the form of case reports. ATN, AIN, MGN, and mesangial hypercellularity with mesangial IGA deposition have been reported in renal biopsies.[190,193,198,199] An autopsy series described kidney histopathologic changes in 69 scrub typhus patients. About 30% had focal or diffuse GN.[200] ATN was found in all cases, with hemoglobin casts in the distal nephron in 69%. Another dominant finding was the presence of varying degrees of interstitial nephritis. The infiltration was most likely to occur near the corticomedullary junction. After the testicles, the kidneys were the second most common site of vascular alterations, such as hyperplasia of the endothelial cells of the glomerulus, thrombophlebitis with interstitial mononuclear infiltrate, and arteriolar necrosis.

DIAGNOSIS

The classical Weil-Felix test, a nonspecific agglutination test that detects antirickettsial antibodies, has low sensitivity and has been replaced by newer methods. Serologic assays such as indirect fluorescent antibody (IFA)- and enzyme-linked immunosorbent assay (ELISA)-based methods are the mainstay of diagnosis. PCR exhibits a higher sensitivity, enabling its utility in the timely identification of diseases at an early stage.[201] Biopsy of eschar or rash can show typical intense lymphohistiocytic vasculitis and thrombosis, along with focal areas of cutaneous necrosis. Culture can be laborious and time intensive due to the obligate intracellular nature of *O. tsutsugamushi* and is available only in specialized centers.

MANAGEMENT

Azithromycin and doxycycline are widely used efficacious antimicrobial agents for the treatment of scrub typhus. Tetracycline and rifampicin have also demonstrated efficacy. Combination therapy with intravenous doxycycline and azithromycin shows superior therapeutic efficacy compared with the use of either drug as a monotherapy.[202] In a trial comparing monotherapy with combination, complications requiring organ support including dialysis were fewer by day 7 in the combination therapy group, with more frequent resolution of hepatic and renal involvement.

PROGNOSIS

The case fatality rates due to scrub typhus vary, with a median mortality rate of 6.0% and 1.4% in untreated and treated patients, respectively. The fatality rate can be as high as 25% in those with MODS.[203] Patients with AKI are at higher risk of mortality.[191,204,205] Higher rates have also been reported in patients who have oliguria and those who undergo dialysis.[190] There has been a noticeable decline in fatality rates associated with scrub typhus infection in recent years, from 12% to 16% to 1.79%.[191,205–207] This improvement could be linked to the increased awareness resulting in rapid diagnoses and prompt commencement of treatment. Persistent urinary abnormalities or reduced GFR can be seen in a minority of patients.[190,193]

BACTERIAL AND SPROCHETAL INFECTIONS

DIARRHEAL ILLNESS

Diarrheal diseases, usually caused by waterborne infections, rank second after respiratory tract infections as the leading cause of infection-related deaths worldwide.[208] The burden is concentrated in low-income and lower-middle-income countries, especially in the pediatric population. Traveler's diarrhea is reported to affect more than 50% of individuals who travel from developed nations to developing nations.[209] Diarrhea can be toxin-induced (*Vibrio cholera,* enteroaggregative/enterotoxigenic *Escherichia coli, Clostridium perfringens, Bacillus cereus, Staphylococcus aureus, Cryptosporidium* spp., *Microsporidia,* and *Cyclospora* spp.); inflammatory (*Shigella* spp., *Salmonella* spp., *Campylobacter jejuni,* enterohemorrhagic or enteroinvasive *E. coli, Klebsiella oxytoca,* and *Entamoeba histolytica*), or enteropenetrative (*Salmonella typhi* and *Yersinia enterocolitica*). Viruses cause diarrhea through a combination of the first two mechanisms. Rotavirus gastroenteritis is typically more common among children younger than the age of 2, while norovirus, adenovirus, and bacterial gastroenteritis are more commonly observed in older individuals.[210,211]

AKI due to prerenal causes, ATN, or HUS is one of the most serious consequences of infectious diarrhea and has been linked to increased mortality rates, extended hospital stays, elevated expenses, and the potential development of CKD. In a study, 1 in 10 adults hospitalized with diarrheal illness experienced AKI, and the occurrence of AKI was associated with a fivefold rise in mortality.[212] Volume depletion leading to AKI is frequent among children in rural areas and urban slums. The frequency of AKI and overall mortality attributed to diarrhea have declined in recent years owing to advancements in sanitation practices, increased adoption of oral rehydration solutions, and the implementation of childhood rotavirus vaccination.

As the frequency of ATN has declined, diarrhea-associated HUS (D + HUS) has become an important cause of AKI in young children. D + HUS is predominantly linked to the presence of Shiga toxin–producing *E. coli* (STEC) and manifests with TMA, characterized by microangiopathic hemolytic anemia (MAHA), thrombocytopenia, and renal dysfunction. The occurrence of STEC-HUS is most prevalent during the period spanning from June to September. Outbreaks have been identified in different parts of the world, specifically in France and Argentina involving STEC O157, associated with consumption of contaminated ground beef and raw milk cheese, and STEC O104, associated with contaminated fenugreek sprouts.[213–215]

Approximately 40% to 60% of children diagnosed with D + HUS require KRT. A meta-analysis showed that 0% to 30% (pooled incidence of 12% [95% CI, 10%–15%]) of individuals with STEC-HUS died or developed end-stage kidney failure (ESKF), whereas 64% were left with abnormal GFR, proteinuria, or hypertension.[216] Residual proteinuria was seen in about 31% of patients.

Supportive therapy focusing on maintaining fluid balance is the main strategy for treating STEC-HUS. Eculizumab has been used for treating those with neurologic involvement, but evidence supporting its use in kidney disease in STEC-HUS is lacking. A retrospective matched cohort study involving 18 children failed to show a benefit of eculizumab on renal and extrarenal outcomes in STEC-HUS.[217] In another randomized, placebo-controlled trial among pediatric patients with STEC-HUS, eculizumab treatment did not correlate with improved renal outcomes during the acute phase of the disease, but those treated with eculizumab had fewer longer-term sequelae when followed up for 1 year.[218] A systematic review of 14 observational studies failed to find a positive effect of eculizumab on medium to long-term outcomes.[219] Recurrent diarrheal diseases increase the risk of CKD either through multiple episodes of AKI or by worsening other underlying conditions, such as nephrolithiasis.

ENTERIC FEVER

Enteric fever, characterized by fever, abdominal pain, and diarrhea caused by *Salmonella typhi* and *Salmonella paratyphi,* is common throughout the tropics. It can be complicated by gastrointestinal hemorrhage and perforation, pneumonia, encephalitis pancreatitis, and AKI. AKI is seen most frequently in the setting of dehydration but can also be associated with intravascular hemolysis in those with glucose 6-phosphate dehydrogenase (G6PD) deficiency, acute GN, AIN, and atypical HUS.[220–223] Nephrotic syndrome has been described in those with concurrent *Schistosoma mansoni* infection (in endemic areas) when *Salmonella* invades the systemic circulation of the host and attach to the tegument of the adult *Schistosoma* parasites.[224] Coinfection of *Salmonella* and *Schistosoma* is of particular concern in the endemic areas of sub-Saharan Africa.[225] Those with kidney involvement may exhibit hypocomplementemia, reduced IgG and IgM, and elevated IgA levels.

The primary therapeutic modalities involve using fluoroquinolones, third-generation cephalosporins, and azithromycin. Managing enteric fever has become challenging due to the emergence of multi–drug-resistant organisms. Kidney involvement is managed by supportive measures and does not require specific therapy.

CHOLERA

Infection with the bacterium *Vibrio cholerae* is endemic in more than 40 countries and is responsible for outbreaks of profound secretory diarrhea, severe emesis, and dehydration.[226] The global annual incidence of the infection is estimated at 1.3 to 4 million, with 21,000 to 143,000 deaths. A large proportion develop dyselectrolytemias, such as hyponatremia, hypernatremia, and hypokalemia. Oligoanuric AKI secondary to ATN was common in the past but has become less frequent following the widespread availability and implementation of oral rehydration strategies.[227,228] Cholera can be life-threatening, and its effective management is heavily contingent on the timely initiation of fluid resuscitation and treatment with one of the effective antibiotics like macrolides, fluoroquinolones, or tetracyclines. A new oral cholera vaccine has helped reduce massive outbreaks in many countries in sub-Saharan Africa.[229]

LEPTOSPIROSIS

Leptospirosis, the most common zoonosis worldwide, is caused by the highly motile aerobic spirochetes of the genus *Leptospira,* of which there are 64 recognized species, split into two clades.[230] Twelve species belonging to subclade P1 are considered pathogenic. More than 160 mammalian species including dogs, pigs, horses, and other cattle are natural *Leptospira* reservoirs. The leptospires colonize the renal tubules and are shed in the urine. Small rodents (e.g., rats), the most important reservoirs that maintain the chain of transmission, get infected in utero and shed leptospires throughout their lives.[231] The leptospires can survive for days in contaminated water and soil. Humans are accidental hosts, acquiring illness through inadvertent contact with animals or environmental exposure when the spirochete enters the circulation through cuts or abraded skin, mucous membranes, or conjunctivae. Risk factors for acquiring the infection include outdoor activities in water bodies like rafting, swimming in freshwater bodies of water, or occupations such as farming, sewer and abattoir work, ranching, and landscaping, especially if done barefoot.

EPIDEMIOLOGY

Leptospirosis is seen worldwide, with the highest prevalence in South and Southeast Asia, Oceania, the Caribbean, sub-Saharan Africa, and Latin America.[232] According to a systematic review, more than 1 million individuals are affected worldwide, and 60,000 die every year.[233] Outbreaks tend to develop in regions characterized by endemicity where housing and sanitary infrastructure are inadequate, particularly in the aftermath of intense rainfall or flooding. Global warming is leading to the expansion of the geographic areas optimal for survival and the spread of leptospires. The reemergence of leptospirosis in Tanzanian farmers was documented in 2022, with 20 verified symptomatic cases and three fatalities.[234] According to the European Climate and Health Observatory, the higher temperatures and rainfall due to climate change will likely increase disease risk and spread to Europe.[235] Nonimmune travelers to endemic areas are especially vulnerable. In fact, the yearly incidence of travel-related leptospirosis in Southeast Asia is greater than that of endemic cases (1.78 vs. 0.06 cases per 100,000 population).[236]

CLINICAL FEATURES

The clinical manifestations of leptospirosis range from an asymptomatic or mild disease to a severe life-threatening illness. The disease typically presents in two phases—an initial bacteremic phase and a later immune phase. Common clinical presentations include fever, rigors, myalgias, nausea, vomiting, diarrhea, arthralgias, and abdominal pain. Jaundice, hemorrhagic manifestations, and conjunctival hyperemia are the distinguishing features of *Leptospira* infection. Organ-specific complications include AKI, liver dysfunction, respiratory failure, uveitis, myocarditis, meningitis, and rhabdomyolysis. Weil disease is the name given to the most severe presentation of leptospirosis, characterized by jaundice, AKI, and bleeding diathesis.[231]

Kidney involvement is nearly universal and ranges from asymptomatic urine abnormalities to severe AKI. Abnormal urinalysis (leukocyturia, hematuria, and bile and granular casts); tubular function abnormalities (glucosuria, bicarbonaturia, uricosuria); and dyselectrolytemia (hypokalemia and hypomagnesemia) are common. AKI develops in 10% to 60% of cases[237] and can be oliguric or nonoliguric. Although more frequent with severe disease, AKI can be seen in those with mild anicteric disease as well.[238] Arrhythmia, oliguria, jaundice, crackles, hyperbilirubinemia, raised direct bilirubin level, increased activated prothrombin time, and leukocytosis were recognized as independent risk factors for the development of AKI.[239]

In recent years, the possibility of endemic leptospirosis leading to the development of CKD, especially in agricultural workers, has been proposed. A longitudinal study from Sri Lanka demonstrated that 9% of patients who had AKI due to leptospirosis developed stage 3 CKD after 1 year.[240] A population-based study conducted in Taiwan with a median follow-up of 8 years indicates that individuals who experienced AKI following infection with leptospirosis were at an increased risk of developing CKD.[241] The risk was directly proportional to the severity of AKI. The hazard ratio (HR) for developing new-onset CKD in AKI requiring KRT compared with those without AKI group and in those with AKI without KRT compared with the non-AKI group was 7.79 (CI 5.29–11.47, $P < 0.001$) and 6.29 (CI 4.86–8.15, $P < 0.001$), respectively, after adjustments for age, gender, and comorbid conditions.

DIAGNOSIS

Diagnosis is based on clinical suspicion and confirmed by serological tests (IgG and IgM antibodies) or PCR in biological fluids. The latter is most likely to be positive in the acute bacteremic phase, and the former in the immune phase. Confirmation by serologic testing requires the demonstration of rising titers on repeat testing. RT-PCR exhibits higher sensitivity and specificity compared with regular PCR.[242] Other tests include antigen detection, blood smears, and *Leptospira* culture.

The clinical Thai-Lepto-on-admission probability (THAI-LEPTO) score was developed by the Thai-Lepto AKI study group for presumptive early diagnosis of leptospirosis in those presenting with a high pretest probability of the disease on the basis of clinical suspicion. The scoring system has seven variables with weighted scores: low hemoglobin (3), hypokalemia with hyponatremia (3), hypotension (3), jaundice (2), muscle pain (2), AKI (1.5), and neutrophilia (1). The test has positive and negative predictive values of 87% and 58%, respectively, with a cutoff at 4.[243] Kidney injury molecule-1 (KIM-1) has been identified as a promising marker for early prediction/early diagnosis of leptospirosis-associated AKI, but further studies are warranted.[244]

PATHOLOGY

The most common histologic findings in leptospirosis are ATN and AIN.[243-246] A biopsy series of patients with leptospirosis showed either mesangial proliferation or no remarkable change in the glomeruli.[247] Tubular degeneration and interstitial edema with cellular infiltrates of mononuclear and few eosinophils were seen in the tubulointerstitial compartment. There was deposition of C3 in the arteriolar wall in IF studies. Leptospiral antigen deposition was detected in the interstitium and tubular cells. Vasculitis has been reported rarely.[245]

PATHOGENESIS

Kidney involvement is linked to direct invasion of the organ by the spirochete leading to cytopathic or immunologically mediated changes. Tubular and electrolyte complications are mostly related to the decrease in the activity of sodium-hydrogen exchanger isoform 3 (NHE3) in the apical membrane of the proximal tubule and decrease in the Na^+/K^+-ATPase on the basolateral aspect.[248]

Hyponatremia is caused by increased urinary sodium loss, cellular sodium efflux due to Na^+/K^+-ATPase abnormalities, high levels of antidiuretic hormone (ADH), or resetting of the osmoreceptor.[249] Downregulation of the sodium-potassium-2-chloride co-transporter (NKCC2) located on the medullary thick ascending limb of the loop of Henle also contributes to urinary losses of sodium and potassium.[250] *Leptospira*-induced tubular damage renders the intraluminal electric charge positive, leading to reduced magnesium and calcium reabsorption.[251]

The primary antigen of the bacteria, LipL32, adheres to the extracellular matrix of the tubular cells, interacts with Toll-like receptor 2, and induces the migration of inflammatory cells into the renal interstitium.[252] The *Leptospira* outer membrane protein (OMP) can also induce host cellular damage by activating the complement system.

Activation of innate immunity via host Toll-like receptor-dependent cascades leads to the activation of nuclear factor–κB (NF-κB), kinases, and cytokines. A massive release of inflammatory cytokines leads to the development of a "cytokine storm."[253] Other factors, such as hypovolemia, jaundice, and rhabdomyolysis, associated with the acute illness, also contribute to the development of AKI.

Evidence has accumulated around the potential of leptospiral infection in the kidneys to induce renal inflammation and subsequent kidney damage. In cases where the kidney fails to recover fully and the infection persists, there is a possibility of the development of renal fibrosis.[254] In vitro cell culture and in vivo animal studies have shown that the involvement of TGF-β1/Smad-Dependent Pathway and Wnt/-catenin pathways are associated with leptospirosis-related kidney fibrosis.[255] In immunohistochemical and quantitative real-time PCR experiments, *Leptospira*-infected C57BL/6J mice produced renal fibrosis with persistent carriage of *L. interrogans*.[256] Chou and colleagues[254] studied the transcriptome profiles of kidneys of mice with adenine-induced and chronically *Leptospira*-infected kidneys and found evidence of chronic inflammation, activated T-helper 17 immune responses, and a high-level expression of indoleamine 2,3-dioxygenase 1 protein, suggesting the risk of progressive kidney injury.

MANAGEMENT

Treatment entails a combination of supportive care and antibiotics. The latter can shorten the duration of disease and hospitalization and reduce AKI risk but does not affect overall mortality. Mild disease should be treated with oral doxycycline or azithromycin, while severe disease requires intravenous doxycycline, ceftriaxone, or cefotaxime. An acute inflammatory response to the clearance of spirochetes, characterized by fever, rigors, and hypotension (Jarisch-Herxheimer reaction), may develop in about one-fifth of cases and can lead to worsening of preexisting AKI or new-onset AKI.[257] Steroids and plasmapheresis have been proposed as adjunct therapies in such cases. In severe disease, supportive care with KRT, ventilatory support, and administration of blood products is often required.

Leptospiral AKI has a mortality rate of 10% to 22%, and severe AKI with multiple organ failure has a higher mortality rate.[237,239] A study by Andrade and colleagues[258] showed that daily dialysis reduced mortality to 16.7% in patients with leptospirosis who were on mechanical ventilation, compared with 66.7% in those who were dialyzed on alternate days.

PARASITIC INFECTIONS

MALARIA

Malaria, the most prevalent vector-borne parasitic infection worldwide, is caused by 5 of the more than 200 known species of the protozoan *Plasmodium: P. falciparum, P. vivax, P. ovale, P. malariae,* and *P. knowlesi. Plasmodium* grows in Culex and Anopheles mosquito hosts and is transmitted to humans when the female Anopheles injects the parasite into the bloodstream during a blood meal. These species predominate in warmer regions closer to the equator.

According to the WHO Malaria Report, 249 million malaria cases were recorded worldwide in 2022, representing an increase of 5 million cases compared with 2021, with 608,000 fatalities. The WHO regions of Southeast Asia, the Eastern Mediterranean, the Western Pacific, and the Americas all report considerable numbers of cases and fatalities. Kidney involvement can be seen in all forms of malaria but is most frequently encountered after *P. falciparum* infection. The epidemiologic spectrum of malaria-associated kidney disease has evolved over the past 30 years. While *P. vivax* was not reported to cause kidney disease in the past, numerous reports have documented the development of a variety of renal lesions including ATN and HUS in recent years. Further, *P. knowlesi,* restricted until recently to causing disease in animals, has emerged as the dominant species causing kidney disease in Southeast Asia.

CLINICAL FEATURES

The disease is dominated by fever, typically accompanied by rigor, headache, malaise, weakness, gastrointestinal discomfort, and muscle pain. Physical findings include pallor, petechiae, jaundice, hepatomegaly, and splenomegaly. Severe cases may be associated with shock, ARDS, AKI, coagulopathy, and hepatic failure.

The spectrum of renal involvement in malaria ranges from asymptomatic urinary abnormalities like proteinuria and hematuria to AKI and, less commonly, nephrotic syndrome.[259–261] Oliguric AKI is the most common presentation, usually accompanied by multiorgan involvement such as jaundice and cerebral malaria. In addition to elevated serum creatinine, hyperbilirubinemia, anemia, and thrombocytopenia are common accompaniments. The global prevalence of AKI ranges from 20% to 50% in cases of malaria requiring hospitalization.[259,261–263] The WHO defines AKI in malaria as a serum creatinine level of 3 mg/dL or a blood urea level of more than 120 mg/dL. These thresholds represent approximately 3-fold elevations compared with standard adult values and 5- to 10-fold for a child. This contrasts with the KDIGO definition of AKI. Consequently, they can only identify advanced, severe AKI, precluding early detection and intervention.[264] In a 25-year study of 671 malaria patients with AKI, *P. falciparum* was the most common cause, followed by *P. vivax* (15.2%) and dual infection (3.57%). KRT was required in 76.6%, and 65% recovered.[263] AKI in malaria has been reported most frequently from South and Southeast Asia and develops more frequently when malaria develops in nonimmune individuals.[265] For example, AKI is reported in 25% to 30% of Europeans with complicated *P. falciparum* malaria, as compared with <5% of those living in endemic areas.[261,266]

Glomerular diseases are uncommon in malaria. Early studies from sub-Saharan Africa in the 1980s and 1990s had documented proteinuria in 46% of cases with *P. malariae* infection, and *P. malariae*–associated immune complex mesangiocapillary glomerulonephritis was reported to be the commonest cause of nephrotic syndrome among the pediatric population in Africa.[267] This condition is not reported in recent literature from Africa. Glomerular lesions have also been reported in *P. falciparum* infection and include IgA nephropathy, MCD, eosinophilic GN, and collapsing glomerulopathy. A recent report from France reported 23 cases of glomerular disease (22 of African ancestry) in association with malaria (22 with *P. falciparum* and 1 with *P. malariae*). Kidney biopsy showed focal segmental glomerular sclerosis in 21 cases (including 18 with collapsing GN) and MCD in two cases. Apolipoprotein L1 high-risk genotype was detected in all seven patients in whom it was tested.[268]

PATHOGENESIS

The mechanisms postulated to explain renal involvement include a combination of hemodynamic perturbations, immune-mediated glomerular injury, and metabolic disturbances.[260,269,270] Parasitized red blood cells (RBCs) exhibit enhanced cytoadherence properties, mediated by increased expression of *P. falciparum* erythrocyte membrane protein (pfEMP), with which they adhere to surrounding healthy erythrocytes, platelets, and the capillary endothelial lining. This process leads to the formation of intravascular clusters and rosettes, leading to tissue hypoxia. A high level of endothelial activation markers, namely soluble intercellular adhesion molecule-1 (sICAM-1), soluble vascular cell adhesion molecule-1 (sVCAM-1), and soluble fms-like tyrosine kinase-1 (sFlt-1), as well as immune activation markers—soluble tumor necrosis factor receptor-1 (sTNFR1), soluble triggering receptor expressed on myelocytes (sTREM-1), chitinase-3-like protein 1 (CHI3L1), and interleukin-8—have been documented in severe malaria.[271,272] CHI3L1 and sTREM-1 have been identified as indicators of disease severity and mortality in malarial AKI.[272]

Additional factors contributing to AKI include hypovolemia due to the insensible loss of fluids, insufficient oral intake, intractable vomiting, hemoglobinuria, myoglobinuria, and immune activation directed against the surface antigens on the parasitized cells. Monocyte activation induces the release of proinflammatory cytokines, and the activation of type 1 T-helper (Th1) cells results in the proliferation of B lymphocytes, leading to generation of autoantibodies and development of immune-mediated GN.[260,269,270,273] Finally, antimalarial drugs like mefloquine and quinine can cause TMA and AKI.[274] The gametocidal Primaquine can cause hemolysis and AKI in individuals with G6PD deficiency.[275] Fig. 59.5 shows the pathogenesis of kidney injury in malaria.

PATHOLOGY

ATN and TMA are the most common histopathologic findings. Tubular cells show vacuolation and are shed into the lumen, whereas glomerular and peritubular capillaries may show RBC rosettes.[276] Jaundice, myoglobinuria, and hemoglobinuria may lead to bile acid and pigment cast nephropathy.[277] Special stains often reveal the presence of iron pigment in the tubules. AIN and GN are seen

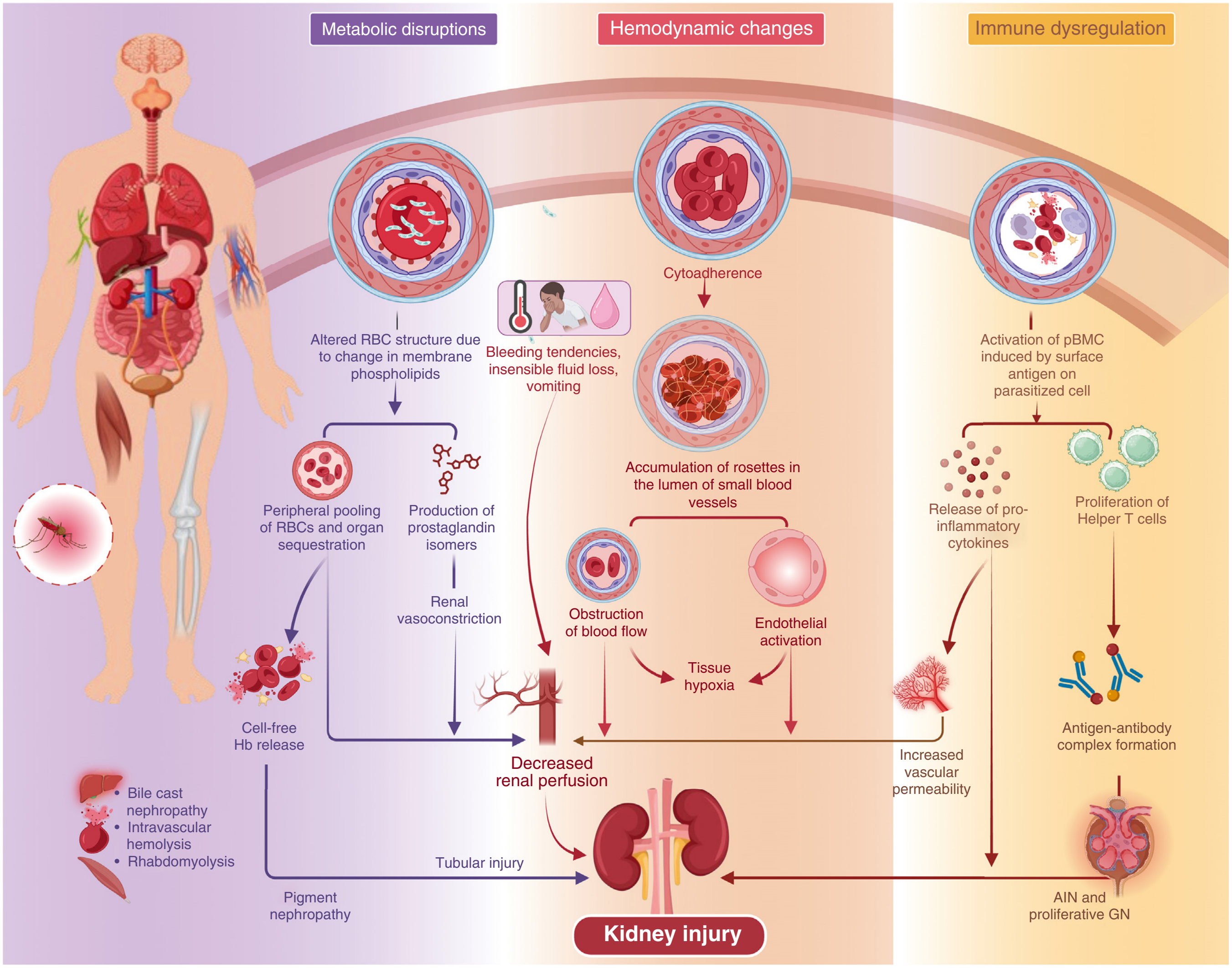

Fig. 59.5 Pathogenesis of kidney injury in malaria. *AIN,* Acute interstitial nephritis; *GN,* glomerulonephritis; *Hb,* hemoglobulin; *PMBC,* peripheral blood mononuclear cells; *RBCs,* red blood cells.

less frequently.[269] Acute renal cortical necrosis has been infrequently reported in *P. vivax* infection.[278,279] Fig. 59.6 shows kidney biopsy of a patient with malaria.

Human malaria parasites including *P. falciparum, P. vivax,* and *P. knowlesi* in both mono and mixed species infections were detected in kidney tissue by a retrospective analysis of renal biopsies taken from malaria cases with AKI. The observation of microvascular injury in a significant proportion of cases indicates the potential involvement of the vascular system.[280]

DIAGNOSIS

Malaria is diagnosed by examining stained blood smears from capillary (finger prick) blood samples to visualize the parasites, rapid diagnostic tests (RDTs) to detect antigens or antibodies in the circulation, and nucleic acid assays. PCR can help with the confirmation of species and identification of drug-resistant mutations. Some studies have evaluated the diagnostic performance of biomarkers in predicting the occurrence of AKI due to malaria. In a small study, KIM-1 and neutrophil gelatinase–associated lipocalin (NGAL) showed good predictive performance. Urinary NGAL had a positive predictive value (PPV) of 1.00 (95% CI 0.54–1.00) and a negative predictive value (NPV) of 1.00 (95% CI 0.89–1.00) as a marker of AKI in malaria.[281]

MANAGEMENT

Treatment of malaria is based on timely administration of antimalarial therapy and the provision of supportive measures such as volume resuscitation, correction of hypoglycemia, dyselectrolytemia, acid-base disorders, and KRT if indicated.

All patients with *P. falciparum* malaria should be treated with artemisinin-based combination therapy (ACT) consisting of a combination of an artemisinin derivative, such as artemether or artesunate, and another antimalarial drug, such as lumefantrine, mefloquine, or piperaquine.[282,283] A combination of quinine plus doxycycline, tetracycline, or clindamycin may be used where ACTs are unavailable.[284] Mefloquine alone can be used for people who cannot tolerate ACTs.[282] Acetaminophen showed renoprotection in studies in patients with severe *P. falciparum* malaria, especially in those with extensive intravascular hemolysis.[285]

Because the AKI is often hypercatabolic, KRT may be required early. Studies have reported superior outcomes with hemofiltration compared with peritoneal dialysis in severe malaria.[286] Mortality in patients with malaria-associated AKI requiring hemodialysis is higher in those with higher Acute Physiology and Chronic Health Evaluation (APACHE) II, Glasgow Coma Scale (GCS) Sequential Organ Failure Assessment (SOFA), and MOD Scores and in those requiring inotropic and ventilator support.[287] The administration of artesunate has been associated with the delayed occurrence of hemolysis, leading to the development of AKI after a period exceeding 10 days.[288]

Recent trials of the R21/Matrix-M malaria vaccine have shown high efficacy against clinical malaria in African children. Following a 3-dose schedule, symptomatic cases were reduced by 75% in areas with seasonal malaria transmission, leading to the WHO recommending their use in malaria-endemic areas.[289]

PROGNOSIS

The overall mortality among cases with AKI in malaria is 15% to 50%.[290] Risk factors associated with mortality include late referral, short acute illness, high parasitemia, oliguria, hypotension, severe anemia, hepatitis, and ARDS. Patients with severe malaria and AKI are more susceptible to the development of acute kidney disease and have a higher risk of death after discharge.[291] Survivors of malarial AKI may be at increased risk of developing CKD. In a study from Uganda, the prevalence of acute kidney disease among community children was 6.8%, whereas the corresponding data for individuals who were survivors of severe malaria was 15.6%.[291] AKI is also correlated with long-term neurocognitive impairment and CKD. In

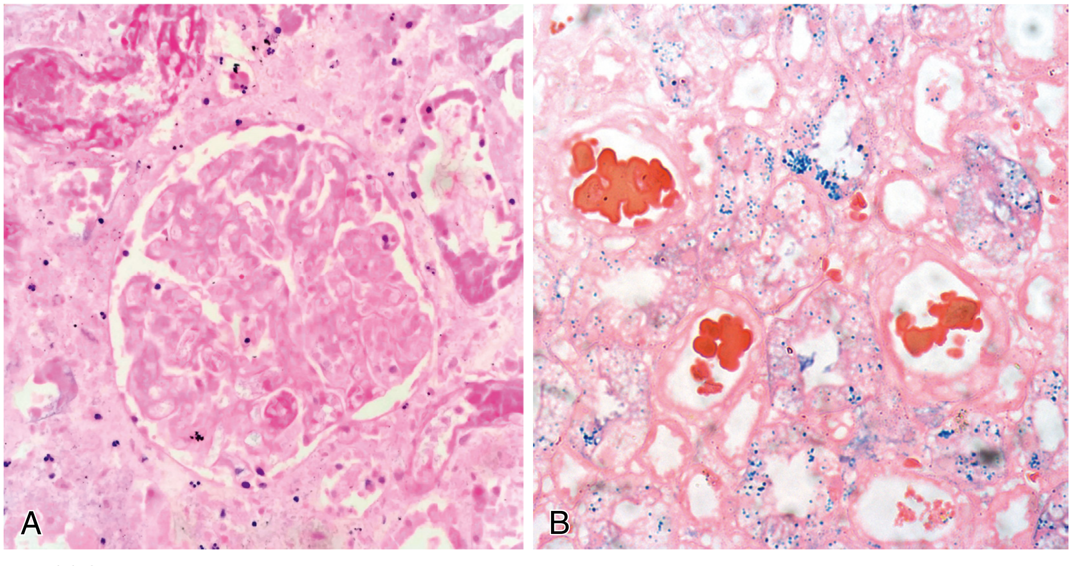

Fig. 59.6 Malaria. (A) Ghost outlines of glomerular and adjoining tubulointerstitial compartment due to acute cortical necrosis in a patient with malarial acute kidney injury (hematoxylin-eosin stain, 40×). (B) Kidney biopsy of a patient with malaria reveals clumps of degenerated red blood cells forming casts with acute tubular injury. Coarse iron pigment is seen in the cytoplasm of proximal tubules (Prussian blue stain, 20×). (Image courtesy Dr. Mahesha Vankalakunti.)

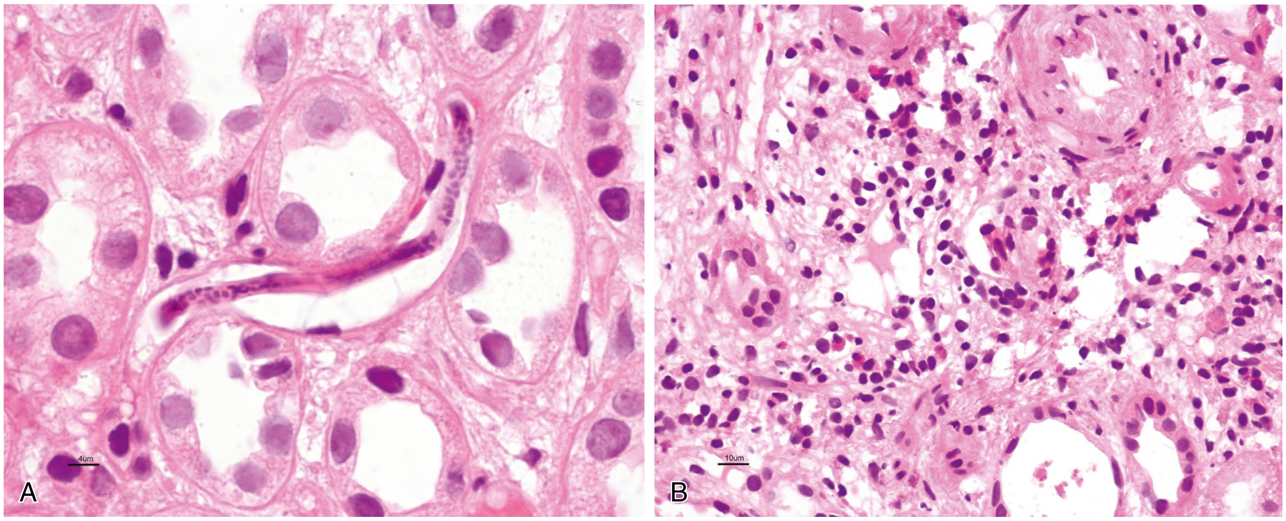

Fig. 59.7 (A) Photomicrograph showing larva of microfilaria in the peritubular capillary lumen in a patient with acute kidney injury (hematoxylin-eosin [H&E] stain, 100×). (B) Acute interstitial edema and inflammatory cell infiltrate rich in eosinophils, as an allergic manifestation of filariasis (H&E stain, 40×). (Image courtesy Dr. Mahesha Vankalakunti.)

a prospective cohort study involving Ugandan children, AKI emerged as a significant risk factor for CKD at the 1-year follow-up, with 7.6% of children with severe malaria-associated AKI developing CKD compared with 2.8% of children without AKI (odds ratio 2.81, 95% CI 1.02, 7.73).[292]

LEISHMANIASIS

Leishmaniasis refers to a collection of parasitic disorders caused by an obligate intracellular protozoan belonging to the genus *Leishmania,* transmitted to humans through the bites of phlebotomine sandflies.[293] Annually, there is an estimated occurrence of 700,000 to 1 million new cases of leishmaniasis.[294]Although it has a worldwide distribution, Asia, Africa, the Americas, and the Mediterranean are endemic regions.[294] Individuals residing in economically disadvantaged regions with inadequate nutrition, forced migration, substandard living conditions, and compromised immune function are at higher risk.

CLINICAL FEATURES

There are four distinct forms of leishmaniasis: mucocutaneous, cutaneous, visceral (VL), and post-kala-azar dermal leishmaniasis (PKDL). VL is associated with a considerable global mortality rate.[295] After an incubation period of a few weeks to several years, the disease manifests insidiously with fever, malaise, weight loss, and splenomegaly, accompanied by pancytopenia and hypergammaglobulinemia.[296] Kidney involvement in VL can manifest as nonspecific proteinuria and hematuria, GN, or AKI. The latter develops in up to 45% of children with VL.[297,298] Impaired urinary concentration is common. Distal renal tubular acidosis has been reported in up to one-third of cases.[299] One study noted increased excretion of low-molecular-weight proteins in 44% of patients.[300] Renal tubular dysfunction has been observed in American cutaneous leishmaniasis, linked to the dysregulation of key acid-base transporters in the proximal tubule (NHE3) and distal nephron (H+-ATPase and pendrin).[301] Finally, amyloidosis can complicate VL.[302]

PATHOLOGY

AIN is the most frequently reported finding in renal histopathology in VL, followed by proliferative and necrotizing GN, MPGN, mixed cryoglobulinemia, and amyloidosis.[303–307] Autopsy studies showed the presence of Leishmania parasites in kidneys.[308]

PATHOGENESIS

ATN in leishmaniasis may be owing to ischemia caused by the parasite leading to the obliteration of small vessels. Parasitic invasion can trigger an inflammatory response, causing necrotizing tubulitis.[303] Autopsy studies have shown that Leishmania can invade all compartments of the kidney.[305-307] The immune system also actively participates, with immune complex deposition, macrophages, T cells, cytokines, and immunoglobulins all playing important roles in the development of immune-mediated renal damage.[296,297] Glomerular diseases could develop from immune complex deposition due to polyclonal activation of B lymphocytes by the parasite.[303,306,311] Fig. 59.7 shows the kidney biopsy of a patient with filariasis.

DIAGNOSIS

RDTs that detect the recombinant 39 amino acid antigen (rk39) are used to diagnose VL.[312] Lymph node, bone marrow, and spleen aspiration can be done to confirm the diagnosis by demonstrating the organism inside macrophages by Giemsa stain.

MANAGEMENT

Amphotericin B, pentavalent antimonials, paromomycin, and miltefosine are effective against leishmania. Antimonial agents have been linked to ATN, and monotherapy with antimonials is not recommended.[313] Distal tubular acidosis, nephrogenic diabetes insipidus, and potassium wasting can develop after amphotericin B therapy.[314] Pentamidine can cause tubular injury, manifesting as hyperkalemia, hypocalcemia, hypomagnesemia, and AKI.[315]

FILARIASIS

Filarial worms and their larvae are parasitic round nematodes encountered in the tropics and cause a group of diseases called filariasis. The infection is transmitted to humans by arthropods, usually mosquitoes. After being released into the bloodstream, the larvae migrate to different tissues of the human body and develop into adult worms.[316] The tissue environment where the worms reside determines the nature of the clinical presentation. Lymphatic filariasis caused by *Wuchereria bancrofti, Brugia malayi,* and *Brugia timori* is the most common form of filariasis and is seen in sub-Saharan Africa, Southeast Asia, South Asia, the Pacific Islands, Latin America, and the Caribbean.[317] Subcutaneous filariasis, caused by *Loa loa, Mansonella streptocerca,* and *Onchocerca volvulus* is encountered in equatorial Africa and South America, whereas serous cavity filariasis caused by *Mansonella perstans* and *Mansonella ozzardi* is seen in Africa and Latin America.[318]

CLINICAL FEATURES

As the name suggests, adult worms in lymphatic filariasis reside in the lymph node sinuses and generate an intense granulomatous reaction leading to blockage of the lymphatics, causing elephantiasis, characterized by woody edema of the draining area, usually the lower limbs. Other manifestations include acute adenolymphangitis, hydrocele, lymphedema, elephantiasis, chyluria, and tropical pulmonary eosinophilia. In the male population, genital involvement can manifest as acute funiculitis and epididymo-orchitis.

In onchocerciasis, the organisms occupy the subcutaneous layer of skin and conjunctiva, leading to the development of subcutaneous nodules, fibrosis, and corneal opacities (river blindness).[319] Loiasis is characterized by attacks of angioedema and cutaneous erythema (Calabar swellings) and subconjunctival injection secondary to migration of the worm in subcutaneous tissue.[320] Mansonella infections present with constitutional symptoms, cutaneous erythematous swellings, and serositis.[321]

Kidney involvement is seen most frequently in lymphatic filariasis. Asymptomatic urinary abnormalities (proteinuria and/or hematuria) are reported in 50% of patients with lymphatic filariasis and 11% to 25% of patients with onchocerciasis and loiasis.[322–324] Nephrotic syndrome is seen in 3% to 5% of cases and is described more commonly in those with polyarthritis and chorioretinitis. Reduction in glomerular filtration is rare.[323,325]

Chyluria (milky urine) resulting from the rupture or regurgitation of lymph from the obstructed and dilated renal lymphatics into the urinary tract can be the first symptom that draws attention to the possibility of lymphatic filariasis in endemic areas.[326,327] Chyluria can be intermittent, may be associated with hematuria and passage of chylous or blood clots, and is quite alarming. Urinalysis shows heavy proteinuria because of the presence of chyle in urine. The proteinuria is often in the nephrotic range but is not associated with glomerular lesions.[328]

The diagnosis is primarily based on clinical suspicion and can be confirmed by the demonstration of microfilaria, filaria antigen, or DNA in circulation. Rarely, microfilariae and/or adult worms are identified in tissue biopsies or cytologic specimens. The patient may show false-positive anti-dsDNA, rheumatoid factor, and autoantibodies against various cytoplasmic proteins. Imaging (ultrasound and lymphoscintigraphy) can be used to detect the presence of adult worms in lymphatic vessels. The diagnosis of kidney disease requires a high index of suspicion. Patients with urinalysis showing heavy proteinuria are often investigated for nephrotic syndrome including kidney biopsy and receive treatment with corticosteroids with a presumed but mistaken diagnosis of MCD if the history of chyluria is not obtained. History of chyluria should always be elicited in endemic areas from those presenting with nephrotic range proteinuria but no edema.[327,329] This will prevent the need for an unnecessary kidney biopsy. Contrast lymphangiography or lymphoscintigraphy is used to visualize the dilated lacteals around the renal pelvis and ureters in those with chyluria. Kidney biopsy may reveal larvae of microfilariae in the glomerular or peritubular capillary lumen.

MANAGEMENT

Diethylcarbamazine is an effective microfilaricidal and macrofilaricidal agent against *W. bancrofti, B. malayi,* and *B. timori*.[330] Doxycycline has macrofilaricidal properties and is often added to the treatment regimen.[331] Mass drug administration is an effective strategy for reducing the bloodborne reservoir of microfilariae to a level that is insufficient for sustained transmission by local mosquito vectors and has led to a remarkable decline in the incidence of filariasis cases following the implementation of the WHO's Global Programme to Eliminate Lymphatic Filariasis.[332] Chyluria refractory to diethylcarbamazine therapy should be managed with low-fat, high-protein diets supplemented with medium-chain triglycerides. Recent reports have suggested that ezetimibe, an inhibitor of intestinal cholesterol absorption, may reduce chyluria.[328,333] In intractable cases, silver nitrate instillation into the urinary tract can obliterate communication with lymphatics and lead to the cure of chyluria.[334]

ECHINOCOCCOSIS

Echinococcosis (hydatid disease), a disease affecting both humans and animals, is caused by various species of the tapeworm *Echinococcus*. The life cycle of this parasite involves dogs as definitive hosts and sheep, cattle, goats, and pigs as intermediate hosts. Humans can also act as intermediate hosts, harboring the metacestode (larval) stage of *Echinococcus* known as "hydatid cysts." Of the four species that infect humans, the disease is caused most frequently by *E Echinococcus granulosus,* followed by *E. multilocularis*. The disease is endemic in central Asia, western China, South America, eastern Africa, and the Mediterranean countries.[335] Prevalence rates are as high as 2% to 9% in endemic regions.

The main risk factors for contracting the disease include contact with canids and involvement in livestock raising. In cases of primary cystic echinococcosis (CE), hydatid cysts may develop in virtually any organ of the human body.[336] The primary infection is asymptomatic, followed by a long latent phase lasting months to years. Approximately 80% of CE patients exhibit involvement of a single organ, typically manifesting as a solitary cyst localized in the liver or lungs.[337] Renal echinococcosis (RE) is less frequent than hepatic and pulmonary disease, constituting approximately 4% of hydatid disease cases.[336–338]

CLINICAL PRESENTATION

RE is usually asymptomatic. Cysts are detected during ultrasound screenings for unrelated reasons. However, some patients may experience a range of symptoms, such as fever, dull flank pain, nausea, vomiting, or lumbar or abdominal masses that resemble tumors.[338–341] In a series of 147 cases, the most common symptom reported was lumbar or lumboabdominal pain in 84% of patients.[340] Hydatiduria, or the presence of hydatid cysts in the urine caused by cyst rupture, was observed in 28% of the cases. Other reported symptoms included fever (70%), lumbar mass (10%), and hematuria (8%). In rare cases, a bladder hydatid cyst may cause urinary retention, urinary frequency, and urgency. Patients may exhibit eosinophilia, pyuria, and hematuria. The most common areas in the kidney for cysts to be located are the polar regions, followed by the mediorenal region.[342] There have been case reports of glomerular lesions with hydatid disease–related nephrotic syndrome, with the histology showing MCD, MGN, MPGN, and amyloidosis.[342-345]

DIAGNOSIS

Imaging modalities such as ultrasound, computed tomography (CT), and magnetic resonance imaging (MRI) play a crucial role in diagnosing hydatid cysts. Ultrasound is particularly effective in detecting key indicators of hydatid disease. The commonest finding is a smooth, round, anechoic cyst, similar to a benign cyst. Additional findings such as hydatid sand, daughter cysts, and floating membranes within the cyst aid in the diagnosis. Ultrasound is recommended for screening in endemic areas and forms the basis for the internationally recognized classification by the WHO expert group and Gharbi classification.[346,347] Contrast-enhanced CT scans offer a more precise and detailed view of the cyst membrane,[348] aid in identifying daughter cysts, and allow the differentiation of hydatid cysts from kidney abscesses and tumors. Differentiating between complex, multivesicular renal hydatid and other conditions, such as hemorrhagic cysts, chronic abscesses, and malignancies, is essential. The hypovascular nature of papillary renal cell carcinoma makes it a particularly important differential. Fig. 59.8 shows gross image of the resected kidney with hydatid cyst.

Several serologic tests have been developed for diagnosis, with IgG ELISA being the most sensitive and specific. Antigen-specific confirmatory tests can supplement the serologic findings. Zmerli and colleagues[333] showed positivity in 80% cases of RE with a combination of indirect hemagglutination and ELISA. PCR tests may play a role in the future. Percutaneous aspiration is used in those with negative serology to look for hydatid membranes, protoscolices, or hooklets but carries the risk of anaphylaxis.

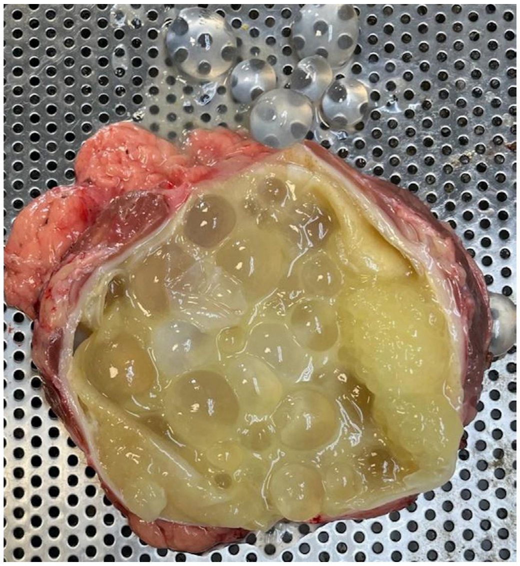

Fig. 59.8 Gross image of the resected kidney (partial nephrectomy) showing a hydatid cyst containing multiple "daughter" cysts. (Image courtesy Dr Abhishek Singh, Muljibhai Patel Urological Hospital, Nadiad, Gujarat, India.)

TREATMENT

Small cysts (<5 mm) can be managed medically with albendazole and monitored by imaging. Alternative approaches include percutaneous aspiration with or without injection of a sclerosant, laparoscopy, and surgery. The latter is the preferred approach for complicated cysts or where many daughter cysts exist. Depending on the remaining viable tissue, surgeons may opt for pericystectomy or nephrectomy.[339–342,349] It is imperative to prevent leakage from the cyst during surgery or aspiration to avoid recurrence or serious anaphylactic reactions.[339,350] Adjuvant treatment with albendazole, mebendazole, or praziquantel is recommended to prevent recurrence.[338,340,349]

FUNGAL INFECTIONS

CANDIDA

Candida species are opportunistic fungal pathogens that colonize the oral cavity, skin, and intestinal tracts of healthy persons. In individuals with compromised immune systems, they have the potential to induce invasive infections. *Candida albicans* is the predominant causative agent of candidemia on a global scale, but there has been a recent trend toward the emergence of non–*albicans Candida* species, such as *C. glabrata, C. parapsilosis, C. kruse,* and *C. tropicalis.*[351] The latter are more likely in people with central venous catheters and those exposed to fluconazole.

The detection of *Candida* species in urine (candiduria) necessitates careful interpretation, ranging from contamination during sample collection to infections of the kidney and collecting system to potentially life-threatening disseminated candidiasis. Risk factors for candiduria include prolonged ICU stay, old age, presence of urinary drainage devices, diabetes, and urinary tract abnormalities.[352] Within the ICU settings, independent risk factors include age older than 65 years, female sex, diabetes mellitus, previous use of antibiotics, undergoing mechanical ventilation, receiving parenteral nutrition, and prolonged hospitalization.[353,354]

PATHOPHYSIOLOGY

Candida urinary tract infections usually begin in the lower urinary tract and spread to the upper urinary system (retrograde infection). This mechanism is facilitated in catheterized individuals who form biofilm, leading to persistence of colonization. Less frequently, they originate elsewhere and seed the kidneys via spread through the bloodstream. For reasons that are not clear, the kidney is the most commonly involved organ in disseminated candidiasis.[355,356] Experimental studies have shown that yeasts adhere to the renal capillary bed using their adherence mechanism, convert to the pseudohyphal form, and migrate to the interstitium.[357]

CLINICAL MANIFESTATIONS

Most individuals with candiduria exhibit no symptoms, with the yeast serving primarily as a colonization agent. Distinguishing between colonization and bladder infection poses a challenge. Some infected individuals can develop dysuria, frequency, and suprapubic pain, while others may be asymptomatic. Candiduria can also be present in those with systemic infection and usually accompanies other signs of disseminated infection.

The clinical presentation of kidney infections varies depending on whether the infection arises secondary to candidemia or ascending bladder infection.[358] Characteristically, candidemia-associated kidney infection is bilateral and presents as microabscesses in the cortex and medulla. Symptoms indicating renal infection include flank/abdominal pain, costovertebral angle, or abdominal tenderness. Renal papillary necrosis or infarction can also develop. Kidney function is seldom affected. Ascending infection is usually subacute to chronic, with the involvement of the renal pelvis and medulla and sparing of the cortex.[356,359] Other manifestations include the development of fungus balls, perinephric abscesses, and less frequently, emphysematous pyelonephritis. The latter is more common among individuals with diabetes and those with structural urinary tract anomalies.[356] Oliguria, strangury, or passage of particulate materials and pneumaturia may suggest the presence of a fungal ball.[360] *Candida* prostatitis can manifest as pain or discomfort in the lower abdomen, pressure behind the pubic symphysis, perineum, or in the anogenital region, and sexual dysfunction.[361] Candidal epididymal-orchitis can manifest with candiduria, swollen tender testicles, and/or scrotal masses.[362]

DIAGNOSIS

Characteristics like yeast count on urine culture, the presence of pseudohyphae, or the presence of pyuria lack discriminatory power between fungal colonization and infection. Pyuria is common in patients with indwelling bladder catheters, which makes it an unreliable indicator of infection. *C. albicans* and most other species appear as yeast cells with a diameter of 4 to 10 mm, exhibiting budding and pseudohyphal formation. However, *C glabrata* is smaller (2–4 mm) and lacks hyphal structures.[363] Demonstration of fungal casts in urine cytology specimens stained with periodic acid–Schiff (PAS) or silver stain, albeit rare, can clinch the diagnosis.

Imaging can identify abnormalities like hydronephrosis, fungal balls, emphysematous pyelonephritis, or perinephric abscesses.[356] Patients manifesting candiduria in conjunction with systemic symptoms or signs should be thoroughly assessed for disseminated infection including imaging studies and blood cultures.

TREATMENT

Antifungal therapy is rarely warranted for asymptomatic candiduria, except in specific high-risk scenarios such as neutropenia, very-low-birth-weight infants, or before a planned urinary tract manipulation in KTRs.[364] Where feasible, it is important to identify and take care of predisposing factors, such as indwelling bladder catheters.

Treatment for symptomatic candiduria involves selecting an appropriate antifungal agent based on the likely Candida species, pending species identification and susceptibility results. Fluconazole (200 mg/day, increased to 400 mg/day for pyelonephritis) is recommended for those not at risk for fluconazole-resistant *Candida,* while intravenous amphotericin B (0.3–0.6 mg/kg/day for 7 days) is preferred for high-risk cases.[364] Flucytosine can be added to those with pyelonephritis. Liposomal preparations of amphotericin are not indicated because they do not penetrate the kidney or reach therapeutic concentrations in urine. The efficacy of echinocandins, particularly micafungin, remains inconclusive, with limited evidence suggesting potential benefits.[365] Experience with voriconazole, posaconazole, and isavuconazole is lacking, with their use considered only in the absence of alternatives.[364] Complications including perinephric abscess, fungus balls, and prostate abscess require a combined surgical and medical approach. Fluconazole is recommended for the treatment of fungus balls until surgical or endoscopic removal is achieved.[360] Irrigation of the urinary tract with amphotericin and endoscopic debulking can hasten recovery in those with fungal balls in the urinary tract. Prostate abscesses, a rare complication, necessitate drainage and antifungal therapy.

MUCORMYCOSIS

Fungi of the order Mucorales, found in soil, decaying matter, and air, are ubiquitous in the environment with minimal intrinsic pathogenicity. However, in immunocompromised individuals such as patients with diabetes, kidney disease, and recipients of solid organ transplants, they can cause devastating and often fatal infections.[366] The range of predisposing factors is continuously expanding including recent additions like COVID-19 infection, trauma, burns, desferrioxamine therapy for iron/aluminum overload, and intravenous drug abuse (particularly in people living with HIV).[366,367] Renal involvement in mucormycosis is mostly observed in the disseminated form, while cases of isolated renal involvement have been sporadically documented.[368,369] Notably, renal involvement has also been reported in immunocompetent individuals.[370] Most infections are caused by *Rhizopus* spp. with *Rhizopus oryzae* and *Rhizopus microsporus* being the most frequently reported organisms, followed by *Apophysomyces elegans, Rhizomucor* spp., *Lichtheimia corymbifera,* and others.[366]

PATHOPHYSIOLOGY

Mucorales are angioinvasive and invade renal blood vessels, resulting in vascular thrombosis leading to ischemia and necrosis of the kidneys. In addition to the renal arteries, these fungi infiltrate the parenchyma, glomeruli, and tubules. The presence of the enzyme ketone reductase supports the growth of the organism in a high-glucose environment. Further, the organism uses iron, making iron-overloaded subjects at increased risk, especially those treated with deferoxamine, a siderophore. In contrast to rhino-orbital-cerebral and pulmonary mucormycosis, both of which likely develop following nasal inhalation of spores, why renal vasculature is particularly susceptible is not clear.

CLINICAL FEATURES

Clinical presentation depends on the nature and extent of organs affected. Kidney involvement is characterized by a mix of fever, abdomen or flank pain and tenderness, gross hematuria, pyuria, oliguria, or AKI.[366,371] In a series of 45 patients with systemic mucormycosis, there were 18 cases with renal involvement, 9 with disseminated illness, and 9 with isolated renal mucormycosis.[371] In 72% of patients, no underlying predisposing risk factor was found. Bilateral involvement was common. Of those with bilateral involvement, the majority (92.3%) of patients presented with anuric AKI. Additional features may indicate the involvement of other organs or sites of origin of the infection.[372]

DIAGNOSIS

The diagnosis of mucormycosis involves identification of organisms in the involved tissue using histopathology. The organism appears as wide, thin-walled, aseptate hyphae showing right-angled branching along with neutrophilic infiltration.[373,374] These fungi can be easily seen using standard hematoxylin-eosin stains and PAS stains. Silver methenamine stains are especially useful. PCR-based techniques can be employed on histologic specimens to further confirm the diagnosis of mucormycosis.[373] Cultures often yield no growth. Contrast-enhanced CT scans often yield characteristic findings, showing enlarged, globular kidneys with complete or partial nonenhancement of renal parenchyma, indicating renal infarction and perinephric stranding. Involvement of adjacent tissues may also be documented. The appearance is characteristic enough to proceed with definitive treatment such as nephrectomy (see later).[375]

TREATMENT

The mainstay of the treatment in mucormycosis is extensive debridement of the involved tissue. Nephrectomy, along with excision of the perirenal tissue, is often required in the case of renal mucormycosis and needs to be bilateral when both kidneys are involved.[366] Predisposing factors such as hyperglycemia, metabolic acidosis, deferoxamine administration, immunosuppressive medications, and neutropenia need to be managed as appropriate. Amphotericin B deoxycholate is the first-line antifungal agent and should be used as an adjunct to surgery. Liposomal preparations are not recommended because they do not penetrate the kidneys or achieve therapeutic concentration in urine.[373] In rare instances, renal mucormycosis has been managed successfully with amphotericin alone without surgery. Posaconazole or isavuconazole is used as a step-down treatment for patients who have responded to amphotericin B or as salvage therapy for those intolerant of this agent. The duration of treatment is not well established but needs to be longer than usual, until there is resolution of the clinical signs and radiologic evidence of active disease. The mortality rate in renal mucormycosis is high, ranging from 44% to 90%.[366,371]

ASPERGILLOSIS

Aspergillosis, a disease caused by various species of the ubiquitous fungus belonging to the genus *Aspergillus,* primarily involves the lungs. However, it can involve other organs, resulting in localized infections.[376–378] The majority of invasive cases are attributed to species belonging to the *Aspergillus fumigatus* complex, followed by *A. flavus, A. niger,* and *A. terreus.*[377] Kidney and urinary tract aspergillosis is a rare but potentially serious infection. Most cases of invasive aspergillosis occur in immunosuppressed patients. Another important risk factor for genitourinary aspergillosis is genitourinary instrumentation.[379] There have been some case reports of isolated renal aspergillosis in immunocompetent individuals.[377–379] Kidney involvement was observed in around 23.4% of cases of invasive aspergillosis in an autopsy series.[380] In another autopsy series of HIV-infected individuals with invasive aspergillosis, kidney involvement was documented in 15%. Rare cases of isolated involvement of kidneys and the urinary tract without evidence of systemic dissemination have been described.[381–383]

PATHOGENESIS

Primary *Aspergillus* infection results from a breach of defenses provided by the mucosa of the respiratory tract and innate immunity. *Aspergillus* can further suppress the immune system by inhibiting macrophage and T-cell functions. Other organs including the urinary tract are involved by dissemination through the bloodstream or spread from adjacent abdominal organs by progression across tissue planes. The risk factors for aspergillosis include poorly controlled diabetes, use of immunosuppressive agents (especially glucocorticoids) and certain biologic agents, liver cirrhosis, hematologic malignancies, organ transplant recipients, genitourinary instrumentation, polymorphisms in *TLR-4* and *IL-1b* genes, and HIV or SARS-CoV2 infections.[377]

Fig. 59.9 shows a gross image of a nephrectomy done due to mucormycosis and renal histopathologic findings in mucormycosis.

CLINICAL PRESENTATION

Kidneys are the most frequently involved segments of the genitourinary system.[377] However, involvement of the prostate, bladder, and ureter has also been observed.[376,377] The presentation can be with various symptoms, none specific to the condition, such as flank pain, fever, hematuria, and systemic symptoms like weight loss and exhaustion.[376,380] Therefore a high suspicion index is required, especially in those with risk factors, to suspect this condition. When present, the passage of particles in urine (fungal balls or bezoars) can draw attention to the condition. Anuria can be a manifestation of bilateral ureteral obstruction due to bezoars.[378]

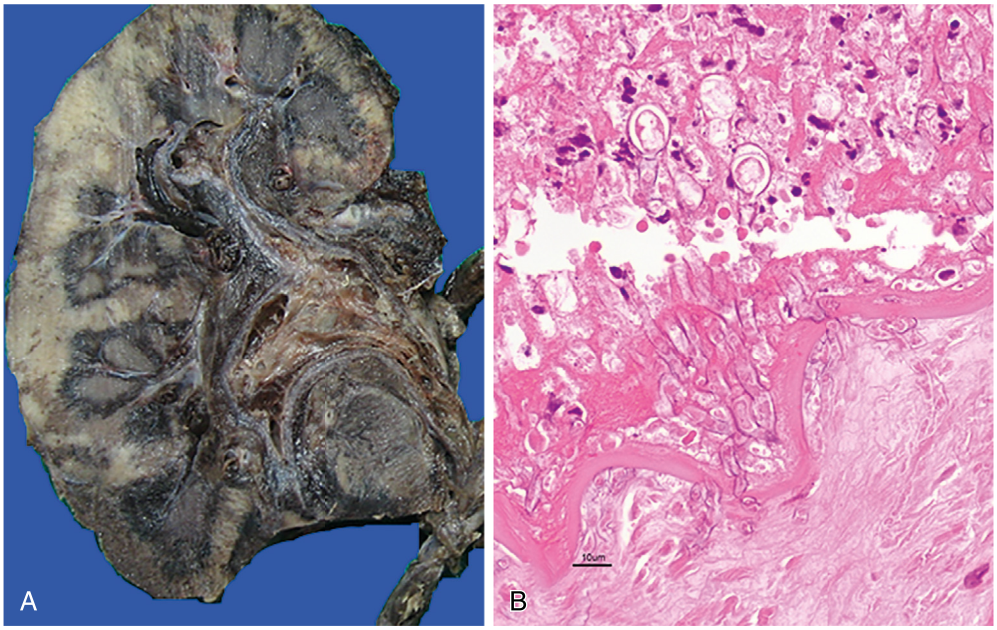

Fig. 59.9 (A) Gross photograph of a nephrectomy specimen showing diffuse hemorrhagic and dark appearance due to mucormycosis. (B) Broad aseptate hyphae of mucormycosis evoking neutrophilic response and invading the internal elastic lamina of the arterial wall (hematoxylin-eosin stain, 20×). (Image courtesy Dr. Mahesha Vankalakunti.)

DIAGNOSIS

The diagnosis of renal aspergillosis depends on detecting *Aspergillus* through either culture or histopathology. CT and MRI are the commonly used methods for identifying, locating, and classifying renal lesions. A systematic review found that histopathology was the primary diagnostic method in more than 70% of the cases of urinary tract aspergillosis.[377] The organisms appear as hyaline, narrow (3 to 6 microns in width), septate hyaline hyphae that diverge at a dichotomous acute angle of 45 degrees.[374] Special stains, such as PAS and Grocott methenamine silver (GMS) are helpful in confirming the diagnosis. Renal abscesses are the most commonly observed finding on CT, characterized as poorly defined isoattenuation or hypoattenuation, depending on the degree of tissue liquefaction.[378,380,384,385] Other manifestations include papillary necrosis, fungal balls, hydronephrosis, and rarely renal infarction due to vascular obstruction from clusters of fungal hyphae.[378,380,384,385] The utility of galactomannan antigen detection, β-D-glucan assay, and whole-blood PCR for the diagnosis of renal aspergillosis is not clear.[377,379,385]

Fig. 59.10 shows contrast-enhanced CT scan of kidneys affected in a patient with mucormycosis.

Fig. 59.10 Contrast-enhanced computed tomography scan showing complete nonenhancement of both kidneys due to renal mucormycosis, except part of the left kidney. The plane between the left kidney and the psoas muscle is obliterated, suggesting invasion.

TREATMENT

The clinical practice guidelines for the diagnosis and management of aspergillosis (2016) by the Infectious Diseases Society of America recommend the use of a combination of medical and surgical therapy for the management of renal aspergillosis.[386] Obstruction of one or both ureters should be treated with decompression using percutaneous and/or endourologic procedures along with local instillation with amphotericin deoxycholate. Percutaneous aspiration or surgical drainage may be needed for parenchymal abscesses. Microwave ablation can be used when surgery is not possible. Voriconazole is most effective for treating parenchymal disease. The efficacy of posaconazole, itraconazole, amphotericin B formulations, and echinocandins in treating renal aspergillosis remains doubtful. Flucytosine has substantial urinary excretion and can be combined with other agents. A systematic review of cases of GU aspergillosis showed a mortality rate of 28.4% in patients with isolated renal aspergillosis.[377]

CONCLUSION

Infections can result in a range of renal abnormalities, including asymptomatic urinary abnormalities, dyselectrolytemis, AKI, and CKD. The epidemiology, presentation, diagnosis, and management are impacted by a confluence of socioeconomic, environmental, and health system-level factors. Poverty, societal behaviors, population density, and inadequate access to essential resources compound the challenge, particularly in tropical regions where climate, culture, and economic conditions further complicate infection prevention efforts. Infection-associated kidney diseases represent a major cause of morbidity and premature mortality in these countries. Ongoing environmental change and biodiversity loss increase the risk of infections and infection-associated kidney diseases worldwide. Prompt diagnosis requires a high threshold of suspicion and access to appropriate diagnostic and therapeutic tools. Addressing neglected tropical diseases necessitates multifaceted public health interventions including water and sanitation initiatives, primary health care provisions, vector control measures, mass drug administration, and vaccination programs. The integration of these efforts demands effective governance, sufficient financial resources, and broad community engagement. Furthermore, ensuring appropriate referral pathways and sustained follow-up protocols are crucial in identifying and managing individuals at risk of kidney diseases.

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